

From Diffusion to Geometry: Detecting the Evolution of Reactive CO₂ Transport in Biosphere 2 LEO

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Abstract

How does comparatively homogeneous mineral parent material acquire the structure and functions of a biologically active soil system? The Landscape Evolution Observatory (LEO) at Biosphere 2 offers a rare opportunity to study this transformation continuously, spatially, and under controlled forcing. The value of the proposed project is therefore not the introduction of a mathematical technique by itself, but the creation of an analytical capability that helps LEO researchers determine when the landscape's functional organization changes, which candidate processes are consistent with that change, and what measurements or experiments should be performed next.

The project integrates experiment, mechanistic modeling, mathematical structure, and uncertainty-aware AI. A multiphase transport–reaction model provides physically testable hypotheses; Buchstaber-style time-series unfolding, affine low-rank approximation, and Grassmann geometry quantify changes in empirical response organization; complex-state and geometric-phase diagnostics summarize relative depth response and wetting–drying path dependence. AI methods support sensor-quality control, multivariable pattern discovery, change detection, surrogate modeling, and held-out event prediction, while physical balances and mathematical invariants constrain what the AI models may claim.

Substantial preliminary work has already audited all 144 basalt CO₂ channels, produced a dated sensor-failure chronology and reproducible analysis window, and shown across 7,489 usable depth-pair months that the 24-hour phase lag cannot be interpreted directly as downward gas diffusion. This establishes data readiness while identifying decisive needs for co-located temperature, pressure, aqueous-carbon, and biological measurements.

Controlled rain events are treated as repeated perturbation experiments. The framework is designed to convert heterogeneous LEO observations into an event atlas, uncertainty-qualified response descriptors, falsifiable mechanism hypotheses, predictive tests, and reproducible software workflows. These products will guide targeted mineralogical, hydrological, geochemical, and biological sampling and connect fast event responses with independently measured slow soil development. Gas-phase CO₂ is used as a sensitive integrating observable, not as a stand-alone definition of soil or proof of a unique mechanism. Transferability to other complex Earth systems will be evaluated rather than assumed. The work is also integrated into the University of Arizona Department of Mathematics course MATH 485/585, creating a research–education pathway for reproducible mathematical modeling of an active Earth-system experiment.

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1 Project Rationale and Objectives

Soil is one of Earth’s most valuable resources. It supports terrestrial life, regulates water and biogeochemical cycles, stores and transforms carbon and nutrients, and sustains human societies. Yet the emergence of soil from mineral parent material remains one of the most difficult Earth-system processes to explain. It is not a single reaction or a monotone trajectory. It arises through interacting physical fragmentation and transport, aqueous and mineral chemistry, pore-structure development, biological colonization, organic-matter dynamics, and feedbacks operating from hours to decades and from mineral surfaces to hillslopes.

Understanding this transformation is consequently important at two levels. Fundamentally, it asks how coupled processes create new system-level function and organization. Practically, it improves the scientific basis for observing, preserving, restoring, and managing soils without assuming that one indicator or one disciplinary model captures the whole system. This project does not claim that an experimental basalt hillslope is equivalent to every natural or agricultural soil; rather, it uses an unusually observable system to identify and test principles that may later be evaluated elsewhere.

LEO provides a rare combination of capabilities for this task. Its similarly constructed basalt hillslopes, dense sensor network, controlled atmosphere and rainfall, and long observational record make it possible to follow an initially comparatively homogeneous granular-basalt substrate as it becomes chemically altered, hydrologically structured, and biologically colonized [1]. Repeated rain events can be used as controlled probes, while direct geochemical, mineralogical, hydrological, and microbial measurements provide independent tests of interpretation. LEO therefore offers something rarely available in Earth science: an evolving landscape in which forcing, response, replication, and long-term history can be studied together.

The central premise is that this opportunity cannot be fully exploited by interpreting measurements one at a time. Individual variables remain indispensable, but the scientific target is the coupled system: which characteristics organize its response, how those characteristics interact, when their relationships change, and whether a model learned in one period can predict a comparable event in another. The project will integrate physical balances, fundamental mathematical structure, statistical inference, and AI-based computation to answer those questions.

Mathematics is used here to make the problem more testable, not more abstract. Conservation laws define physically admissible relationships; unfolding and Grassmann geometry identify reproducible response structure; scale and robustness tests reveal what persists across sensors, events, and preprocessing choices; and geometric observables summarize path dependence without reducing the system to a single concentration. AI complements this layer by handling heterogeneous high-dimensional data, detecting multivariable patterns, constructing uncertainty-aware surrogate models, and generating ranked hypotheses for further testing. AI outputs are not treated as mechanisms merely because they predict well: they must remain consistent with physical constraints, survive held-out-event and held-out-year tests, and lead to experimentally discriminable hypotheses. Leave-one-hillslope-out analysis is exploratory because the three hillslopes provide limited system-level replication.

The project is organized around a two-way learning cycle:

$$\begin{aligned} \text{controlled experiment and observation} &\longrightarrow \text{mathematical structure and physical hypotheses} \\ &\longrightarrow \text{constrained AI models and predictions} \\ &\longrightarrow \text{targeted measurement or new perturbation} \\ &\longrightarrow \text{revised theory and model.} \end{aligned} \tag{1}$$

This cycle is intended to move the analysis beyond unconstrained correlation while acknowledging that mechanism is established only by convergent evidence, not by a single mathematical or machine-learning diagnostic.

Core scientific question.

How does mineral parent material acquire the organization and functions of a living, evolving soil system, and which physical, chemical, and biological principles govern that transformation?

Here “living soil” is an organizing scientific ambition, not a binary label inferred from CO₂. Operational evidence must combine changes in primary and secondary mineral phases, fines and aggregation, organic carbon, pore structure and water retention, nutrient retention and export, biological colonization or function, and the response to repeated forcing. No single component is necessary and sufficient in isolation, and mature fertility is not assumed.

Project objectives. The project will:

1. maintain a dated sensor-failure calendar and build a unified, quality-controlled representation of LEO forcing, fast event responses, and slow mineralogical, hydrological, geochemical, and biological change;
2. identify empirical response structures, characteristic scales, and change points that remain stable across event, sensor, embedding, and preprocessing choices;
3. formulate and test physically admissible alternative explanations for observed changes, explicitly quantifying equifinality and uncertainty;
4. develop constrained AI and reduced-order models that predict held-out event responses within a declared domain of applicability;
5. connect fast dynamical descriptors to independent indicators of basalt weathering and soil development, without defining soil from CO₂ alone;
6. deliver reproducible tools that allow LEO researchers to monitor uncertainty-qualified response descriptors, compare events and hillslopes, and rank competing hypotheses;
7. produce evidence-based improvements to LEO forcing, sensor geometry, co-located validation, calibration, and responsive sampling, and test at least one feasible change prospectively;
8. integrate validated project components into MATH 485/585 to create a reviewed research–education pathway in mathematical modeling;
9. evaluate which parts of the framework transfer to other complex, multiscale Earth-system observatories.

The intended result is therefore tangible: not simply a new index or demonstration of mathematical novelty, but a validated analytical workflow that converts complex observations into quality-controlled response descriptions, falsifiable hypotheses, uncertainty-qualified predictions, and evidence-based priorities for how LEO should deepen or broaden measurement and experimentation.

Primary users and value. For LEO investigators and operators, the framework will provide a living sensor-failure calendar, common event catalogue, response atlas, model-admissibility flags, sampling triggers, and evidence for improving future experiments. For soil, critical-zone, and ecosystem scientists, it will provide a quantitative bridge between fast perturbation response and slow direct indicators of soil development. For students and educators, MATH 485/585 will provide reviewed, reproducible case studies connected to active research. For external observatories and Earth-system modelers, the project will provide documented data schemas, synthetic benchmarks, validation rules, and a transferable perturbation-analysis protocol, together with explicit statements of where transfer fails.

2 Preliminary Research, Data Readiness, and Educational Integration

The project does not begin from an untested concept. A substantial preliminary research effort has already integrated the relevant LEO literature, the physical balance structure of the gas–water–basalt system, Buchstaber’s unfolding framework, geometric-phase methods, and exploratory analysis of the available sensor records. This preparatory work was undertaken proactively to determine whether the proposed research questions are supported by data of sufficient quality and to identify where the present observatory design limits inference.

Sensor-quality audit and failure calendar. A sensor-by-sensor forensic analysis has already been completed for the 144 subsurface basalt CO₂ channels across the three hillslopes. The audit uses explicit criteria for missingness, instrument error codes and implausible values, abrupt transitions into characteristic high-value bands, loss of diurnal amplitude, prolonged constant behavior, excessive fragmentation, disagreement between amplitude- and spectrum-based liveness, and facility-wide gaps. The transition detector was checked against an independently prepared West-slope baseline: 42 of 48 classifications agreed and the median discrepancy in estimated transition date was four days.

The current audit snapshot classifies 46 of 144 channels as recently alive, 13 as intermittent, 81 as dead, and 4 as never reliably commissioned. The full three-slope chronology records an estimated transition date and supporting pre/post statistics for every channel in `data/failure_chronology_all.csv`; monthly state histories are retained in `data/liveness_matrix.json` and `data/liveness_long.csv`. Two common data-acquisition gaps in the principal 2017–2018 window were also isolated, including a 16.5-hour gap on 2017-11-02/03 and a second gap of approximately ten hours. The audit preserves raw data and converts all decisions into dated, reversible analysis masks.

This *sensor failure calendar* is itself a tangible result. On the scientific side, it supplies clean analysis-ready intervals with provenance and exposes survivorship bias: the sensors that remain usable are not a random sample of the hillslopes. On the operational side, it provides LEO investigators and Biosphere 2 leadership with dated evidence for verification, recalibration, replacement, redundancy, metadata correction, or revised maintenance. It should remain a living observatory product updated after calibration, repair, or newly detected inconsistency rather than a one-time appendix.

A defensible analysis window. The quality work identified 2017-10-01 through 2018-11-30 as the longest interval in which borehole R4_C-4 retains a full depth profile on all three hillslopes simultaneously. Within this fixed 14-month interval, 110 of 144 sensors are alive for at least 12 months and eight are clean for all 14 months. This is a survivorship rule, not evidence that R4_C-4 is a physical reference, and approximately one annual cycle cannot separate seasonal from secular change. The project therefore begins with a reproducible data window while explicitly carrying its selection bias and temporal limitation.

Preliminary physical findings and hypothesis generation. The preliminary work has already rejected one tempting but unsafe interpretation. Across the fixed window and a 2012–2026 history totaling 7,489 usable depth-pair months, the 24-hour D1-to-D2 response shows several hours of phase lag but far less amplitude damping than a downward diffusive CO₂ wave requires; D2-to-D3 generally damps more strongly. In the fixed window, only 1 of 35 admissible D1-to-D2 pairs is wave-like under the amplitude–phase consistency test. Consequently, diurnal phase must not be inverted directly as gas diffusivity. The evidence is more consistent with a contribution from locally generated or thermally organized subsurface dynamics, but it does not determine abiotic versus biological origin.

This negative result is scientifically productive because it identifies the decisive measurements missing from the analyzed array: co-located depth-resolved temperature, pore-water DIC/alkalinity, gas pressure, phase-exchange constraints, and direct biological or respiration observations. It also exposed disputed conversion from depth identifiers to absolute centimeters, so absolute diffusivity should not be reported until installation metadata are reconciled. Thus the preliminary analysis has already done what the full project promises: cleaned the evidence base, rejected an attractive but invalid inversion, revealed nontrivial response structure, and translated uncertainty into concrete priorities for deeper measurement and experimental improvement.

Educational and broader scientific impact. The work is not being developed “for the shelf.” It is deeply integrated into MATH 485/585, *Mathematical Modeling*, in the Department of Mathematics at the University of Arizona. LEO provides the course with a scientifically consequential case study in conservation laws, inverse problems, dynamical systems, data quality, uncertainty, numerical computation, and responsible use of AI. In return, the course creates a structured environment for reproducible analyses, independent checks, student research training, and the generation of new testable questions for the observatory.

This educational integration broadens the impact beyond one manuscript or software package. It connects mathematics students with ecologists and Earth scientists, trains researchers to distinguish mathematical identifiability from physical mechanism, and produces documented examples that can support future teaching, outreach, and interdisciplinary research. Course-derived results enter the scientific workflow only after faculty review, reproducibility checks, and the same validation standards applied to the research team.

3 Project Architecture: Three Phases and Decision Gates

The project is organized as three cumulative phases. The phases are not labels for increasingly elaborate mathematics; each ends with a decision about whether the evidence is strong enough to justify the next investment in analysis or experimentation.

Phase I—establish a trustworthy perturbation-response observable. Existing data and repeated rain events are used to determine where the sensor record, boundary conditions, and reduced physical model support reproducible depth-resolved amplitude, phase, apparent transport-exchange, and path descriptors. *Decision Gate I:* proceed only for windows, columns, and event classes that pass quality, coherence, identifiability, and robustness tests.

Phase II—connect fast response, slow soil development, mathematics, and AI. Validated events are analyzed with centered unfolding, Grassmann and projective geometry, independent slow-state observations, model comparison, and uncertainty-aware prediction. The purpose is to test whether response organization changes through time, which explanations remain admissible, and where present observations are insufficient. *Decision Gate II:* retain only findings that survive held-out events, alternative preprocessing and model classes, sensor-removal tests, and comparison with independent mineralogical, hydrological, chemical, structural, and biological evidence.

Phase III—produce and prospectively validate experimental-design knowledge. The information gaps and candidate mechanisms exposed in Phases I–II become an additional scientific and operational result of the work: evidence-based priorities for forcing protocols, sensor depths and locations, co-located measurements, responsive sampling, calibration controls, and, where justified, new manipulations. Biosphere 2 leadership and LEO operators are users and co-evaluators of this research result alongside the scientific team. At least one feasible change is preregistered and tested prospectively. *Decision Gate III:* retain a design change only if it improves identifiability, uncertainty, predictive performance, or discrimination among hypotheses relative to a predeclared standard design.

This architecture closes the apparent gap after Phase I: retrospective analysis is not the endpoint. It supplies evidence for integrated learning in Phase II and for an operational decision about the future design of LEO in Phase III.

4 Scientific Problem and Physical Formulation

The Landscape Evolution Observatory (LEO) at Biosphere 2 was designed as a large-scale, controllable infrastructure for studying the coupled evolution of hydrological, geochemical, biological, and geomorphological processes. It consists of three replicate artificial hillslopes, each with an area of approximately 330 m², an approximately one-meter-thick layer of granular basalt, and an extensive sensor network; precipitation and atmospheric conditions can be experimentally controlled [1]. Long-term, high-frequency measurements in LEO demonstrate the feasibility of synchronized analysis of hydrology, temperature, moisture, and atmospheric variables [2]. A more recent hillslope-scale experiment on granular basalt dissolution used distributed measurements of P_{CO_2} , water content, and total water balance and demonstrated a pronounced connection between hydrological state and CO₂ uptake during basalt weathering [3].

In the present problem, CO₂ concentration is measured over many years both in the pore air of the basalt layer and in the air above the surface. The observed concentration is controlled by at least three competing mechanisms:

- (i) biological generation of CO₂;
- (ii) transport through the pore space, which depends on moisture, porosity, and tortuosity;
- (iii) uptake and geochemical reaction of CO₂ with basalt and the aqueous phase.

Consequently, the same increase in concentration at a given depth may arise from physically different causes. Amplitude changes alone cannot reliably separate source, transport, and sink effects.

The central object of analysis should therefore not be an individual time series $C_j(t)$, but the state of a distributed, multiphase transport–reaction system. If C_g denotes moles of CO₂ per unit gas volume and ε_g is gas-filled porosity, a dimensionally explicit gas-phase balance is

$$\frac{\partial(\varepsilon_g C_g)}{\partial t} = \nabla \cdot (\varepsilon_g D_g \nabla C_g) - \nabla \cdot (\mathbf{j}_{g,\text{adv}}) + S_g - J_{gw}, \quad \mathbf{j}_{g,\text{adv}} = \mathbf{u}_g C_g, \quad (2)$$

where D_g is an effective gas diffusivity, $\mathbf{u}_g$ is a Darcy-scale gas flux, S_g is the net gas-phase source, and J_{gw} is transfer from gas to pore water followed by aqueous speciation and reaction. This balance makes clear that rainfall can change the measured gas signal both by changing gas-filled porosity and connectivity and by transferring CO₂ into water; neither effect is, by itself, proof of mineral consumption.

The reduced equation used below,

$$\frac{\partial C}{\partial t} = \nabla \cdot (D_{\text{eff}} \nabla C) - \nabla \cdot (\mathbf{v}_g C) + S_{\text{bio}} - R_{\text{chem}}, \quad (3)$$

is obtained only after dividing by a locally nearly constant ε_g and absorbing unresolved gas–water exchange into effective source and sink terms. During rapid wetting, when $\partial_t \varepsilon_g$ is appreciable, this reduction must be tested rather than assumed.

For a first local model one may linearize a perturbation about a window-specific reference state C_0 :

$$R_{\text{chem}}(C) - R_{\text{chem}}(C_0) \simeq \kappa_{\text{eff}}(C - C_0). \quad (4)$$

For divergence-free $\mathbf{v}_g$, or after retaining the full conservative advection operator, the reduced transport–reaction operator is

$$\mathcal{L}(t)C = -\nabla \cdot (D_{\text{eff}}(\mathbf{r}, t) \nabla C) + \nabla \cdot (\mathbf{v}_g(\mathbf{r}, t) C) + \kappa_{\text{eff}}(\mathbf{r}, t) C. \quad (5)$$

This operator is an *effective dynamic descriptor*, not the complete physical state of soil. Its inferred drift is evidence for changing transport–exchange organization only when hydrological state, boundary forcing, and sensor response are controlled; mineralogical, aqueous-chemical, and biological measurements are required to attribute that drift to soil formation.

5 Phase I of the Project: Physical and Phase-Based Diagnostics of the LEO Reaction–Diffusion System

The first phase of the proposed project should be deliberately conservative, physically transparent, and directly testable using existing data. At this stage, there is no need to assume in advance the existence of a topological transition, an exceptional point, or even a robust winding number. The more fundamental question is whether long-term spatiotemporal CO_2 measurements can be used to recover reproducible transport and reaction characteristics, and whether the evolution of an entire vertical concentration profile can then be represented as a trajectory of a normalized complex state.

This sequence is a project-level synthesis: first test a reduced reaction–transport model and the meaning of the measured phase; next solve a local inverse problem for a single vertical sensor column; then extend the analysis to other columns; and only after that introduce geometric-phase observables. These physical and projective constructions do not come from the two Buchstaber papers. Those papers supply the separate unfolding, Grassmannian, rank, width, recurrence, and reconstruction layer used later. The search for genuine topological defects is postponed until the robustness of the amplitude–phase field has been demonstrated.

Physical Objective of Phase I At any given time, the measured CO_2 concentration results from the combined action of at least three processes:

$$\text{biological generation} + \text{transport through pore space} + \text{chemical/geochemical sink.} \quad (6)$$

The scalar quantity $C(\mathbf{r}, t)$ is therefore not an unambiguous indicator of the state of the medium. For example, an increase in C at depth may result from an increase in S_{bio} , from a reduction in D_{eff} that slows gas escape, or from a change in κ . Phase I must therefore construct multiple, partially independent observables that constrain and may reject candidate explanations without claiming unique separation.

The minimal governing model is

$$\frac{\partial C}{\partial t} = \nabla \cdot (D_{\text{eff}} \nabla C) - \nabla \cdot (\mathbf{v}_g C) + S_{\text{bio}} - \kappa C. \quad (7)$$

The inclusion of an advective term does not imply that gas advection must dominate. It is retained because rapid changes in water saturation or pressure can temporarily drive the transport regime away from pure diffusion. The first inverse experiment should nevertheless begin with the simplest diffusion–reaction approximation and test its validity against the data.

The controllable structure of LEO is especially important in this context: rainfall can be used as a repeatable external perturbation, while the three hillslopes permit comparison among closely matched replicate systems [1]. Published hillslope-scale LEO experiments also directly connect P_{CO_2} , water content, and basalt weathering, so hydrological state must enter the physical interpretation explicitly [3].

Step I.1: Selection of Reliable Vertical Columns and Data Synchronization The preliminary audit means that Phase I does not begin with blind cleaning. It begins from the dated failure calendar and the fixed 2017–2018 survivorship window, then tests representative reliable columns before expanding to the full admissible set. For each selected column one should:

1. select intervals over which all CO_2 sensors used in the analysis are reliable and continuous;

2. synchronize $C_j(t)$ with temperature, water content or moisture, atmospheric CO_2 , rain/irrigation, and, where available, water-storage data;
3. mark sensor maintenance, calibration changes, missing intervals, and anomalous periods explicitly;
4. place all signals on a common time grid without excessive interpolation;
5. analyze both absolute concentrations and detrended/centered signals, while keeping these two levels of interpretation distinct.

This restriction is essential. Geometric phase is highly sensitive to relative phase relationships between channels; consequently, one poorly functioning sensor can create a spurious phase anomaly. During Phase I, data quality is more important than spatial coverage.

Step I.2: Selection of Physically Relevant Time Scales The relevant frequencies are environmental rather than acoustic. Natural scales are set by environmental forcing and by transport times. At a minimum, one should examine:

1. the diurnal band, $T \sim 24$ h;
2. the semidiurnal band, $T \sim 12$ h;
3. a multi-day response range, approximately $T \sim 2\text{--}10$ days;
4. event-specific rain-response windows;
5. slower scales ranging from several weeks to seasonal and annual variability.

The diurnal and semidiurnal bands remain valuable diagnostic bands, but the preliminary array-wide consistency test has already ruled out direct inversion of the 24-hour D1-to-D2 phase as gas diffusivity. At these frequencies the analysis should compare thermal, local-source, storage, and phase-exchange explanations. Physical inversion is reserved for bands and events that independently pass attenuation-phase, forcing-coherence, boundary, and source tests.

For pure diffusion, harmonic attenuation length is

$$\delta(\omega) = \sqrt{\frac{2D}{\omega}}. \quad (8)$$

With a first-order sink it becomes $\delta_{\text{att}} = 1/\alpha$, where

$$\alpha = \sqrt{\frac{\sqrt{\kappa^2 + \omega^2} + \kappa}{2D}}. \quad (9)$$

Setting the pure-diffusion $\delta = L_z$ gives $\omega = 2D/L_z^2$ and therefore the sinusoidal period $T = 2\pi/\omega = \pi L_z^2/D$. For $D = 10^{-6} \text{ m}^2 \text{ s}^{-1}$ and $L_z = 1$ m, the transient diffusion time L_z^2/D is 11.6 days, whereas the corresponding full-depth harmonic period is about 36 days. These are different conventions, not the same timescale. The analysis should therefore be multi-band, and D and κ should not be fixed a priori.

Step I.3: From Real-Valued Time Series to a Complex Response For each sensor j , a local Fourier component is computed in a sliding window:

$$\hat{C}_j(\omega, t) = A_j(\omega, t) \exp[i\phi_j(\omega, t)]. \quad (10)$$

The amplitude A_j characterizes the strength of the response at the selected frequency, while ϕ_j characterizes the delay relative to the forcing or relative to neighboring depths.

The multichannel complex vector

$$\hat{\mathbf{C}}(\omega, t) = \begin{pmatrix} \hat{C}_1 \\ \hat{C}_2 \\ \vdots \\ \hat{C}_N \end{pmatrix} \quad (11)$$

is normalized as

$$|\Psi(\omega, t)\rangle = \frac{\hat{\mathbf{C}}(\omega, t)}{\sqrt{\sum_j |\hat{C}_j(\omega, t)|^2}}. \quad (12)$$

This provides the appropriate analogue of a Deymier-style wave state: acoustic amplitudes are replaced by the complex Fourier responses of the CO₂ concentration field. This transfer should be understood as a mathematical analogy, not as a claim that the CO₂ field is an acoustic wave. In the work of Deymier and coauthors, a geometric-phase observable was used as a sensing quantity sensitive to changes in the physical medium [18]; in LEO, the object that changes is the transport–reaction operator.

Step I.4: Physical Meaning of the Vertical Phase Consider the local one-dimensional reduced model

$$\frac{\partial C}{\partial t} = DC_{zz} - \kappa C + S(z, t). \quad (13)$$

The inversion below is valid only for a perturbation interval in which D and κ are approximately constant, advection and gas-filled-porosity changes are negligible, the selected harmonic is coherently imposed at a boundary, and the distributed source has negligible power at that harmonic. In a semi-infinite or one-way-propagation approximation, the transfer function relative to the boundary forcing is

$$H(z, \omega) = \frac{\hat{C}(z, \omega)}{\hat{C}(0, \omega)} \propto e^{-qz}, \quad q^2 = \frac{\kappa + i\omega}{D}, \quad q = \alpha + i\beta. \quad (14)$$

For a finite one-metre column, both exponential branches and the actual surface and basal boundary conditions must instead be fitted. If coherence with the chosen forcing is low, $\arg \hat{C}$ is not a transport delay and this inversion is inadmissible. Then

$$\hat{C}(z, \omega) \propto e^{-\alpha z} e^{-i\beta z}, \quad (15)$$

and

$$\phi(z, \omega) = -\beta z + \phi_0. \quad (16)$$

The vertical phase gradient is therefore not merely an abstract phase quantity. It is directly related to transport delay:

$$\beta = -\frac{\partial \phi}{\partial z}. \quad (17)$$

The amplitude gradient gives

$$\alpha = -\frac{\partial}{\partial z} \ln A. \quad (18)$$

Because

$$\alpha^2 - \beta^2 = \frac{\kappa}{D}, \quad 2\alpha\beta = \frac{\omega}{D}, \quad (19)$$

one obtains

$$D = \frac{\omega}{2\alpha\beta}, \quad \kappa = D(\alpha^2 - \beta^2). \quad (20)$$

These relations define candidate *apparent* parameters $D_{\text{app}}(\omega, t)$ and $\kappa_{\text{app}}(\omega, t)$ only when the assumptions stated above pass diagnostics. They are transfer-function summaries, not automatically intrinsic material constants. Admissibility requires positive D_{app} , a physically allowed κ_{app} , stable

phase unwrapping, adequate forcing–response coherence, sufficient amplitude, and consistency among depth intervals.

In the weak-sink limit, $\kappa \approx 0$,

$$\alpha \approx \beta \approx \sqrt{\frac{\omega}{2D}}, \quad (21)$$

and therefore

$$D \approx \frac{\omega}{2(\partial_z \phi)^2}. \quad (22)$$

Thus, the phase gradient provides an independent estimate of transport without requiring the absolute concentration level.

Step I.5: Inverse Problem for a Single Vertical Sensor Triplet Suppose measurements are available at depths

$$z_1 < z_2 < z_3.$$

For each frequency and each sliding window, compute

$$\beta_{12} = -\frac{\phi_2 - \phi_1}{z_2 - z_1}, \quad \beta_{23} = -\frac{\phi_3 - \phi_2}{z_3 - z_2}, \quad (23)$$

$$\alpha_{12} = -\frac{\ln |\hat{C}_2 / \hat{C}_1|}{z_2 - z_1}, \quad \alpha_{23} = -\frac{\ln |\hat{C}_3 / \hat{C}_2|}{z_3 - z_2}. \quad (24)$$

This yields two independent parameter estimates:

$$D_{12} = \frac{\omega}{2\alpha_{12}\beta_{12}}, \quad \kappa_{12} = D_{12}(\alpha_{12}^2 - \beta_{12}^2), \quad (25)$$

$$D_{23} = \frac{\omega}{2\alpha_{23}\beta_{23}}, \quad \kappa_{23} = D_{23}(\alpha_{23}^2 - \beta_{23}^2). \quad (26)$$

The comparison between (D_{12}, κ_{12}) and (D_{23}, κ_{23}) is more than a method of averaging. It is a direct diagnostic of model validity. If the two estimates agree within uncertainty, a simple homogeneous reaction–diffusion approximation is plausible. If they differ systematically, the discrepancy may indicate:

- depth-dependent $D(z)$;
- a localized biological source;
- a localized sink;
- moisture-dependent transport;
- transient advection;
- insufficient stationarity of the selected time window.

Phase I should therefore produce not only estimates of D and κ , but also a quality-of-fit or consistency flag for every analyzed window.

Step I.6: Constraining, Not Uniquely Separating, Candidate Mechanisms Amplitude and phase provide complementary constraints that may reduce—but do not remove—equifinality among source, storage, transport, phase exchange, and reaction. The signatures below are qualitative hypotheses under the restricted reduced model, not unique fingerprints:

Change	Expected signature	Interpretation
$S_{\text{bio}} \uparrow$	increase in $ \hat{C} $, with potentially weaker change in phase gradient	enhanced local generation
$D \downarrow$	increase in $ \Delta\phi $ and transport delay	reduced effective pore connectivity
$\kappa \uparrow$	stronger amplitude attenuation together with a change in phase gradient	enhanced effective sink/reaction

After fitting the reduced model, define only an unresolved balance term

$$S_{\text{res}} = \frac{\partial C}{\partial t} - \nabla \cdot (D \nabla C) + \kappa C + \nabla \cdot (\mathbf{v}_g C). \quad (27)$$

It contains biological production and consumption together with omitted variable gas storage, gas–water exchange, aqueous and lateral transport, boundary-flux error, parameter error, and sensor-scale mismatch. With three depth sensors, direct differentiation is noise amplifying and generally underdetermined; S_{res} should be estimated only through a regularized weak-form or state-space fit with propagated uncertainty. It may be labeled biological only after the other terms are independently constrained.

Step I.7: Using a Rain Pulse as a Controlled Perturbation The most powerful experimental design for Phase I is to compare responses to comparable rain events. For each event, define the event time

$$\tau = t - t_{\text{rain}}, \quad (28)$$

and analyze the chain

$$\text{rain} \longrightarrow \theta_w(\tau) \longrightarrow \{D(\tau), \kappa(\tau)\} \longrightarrow \hat{C}_j(\omega, \tau). \quad (29)$$

Repeatability is critical. If two rain events have similar forcing and comparable antecedent hydrological and thermal states but exhibit different vertical phase lags and different admissible apparent D and κ , the result is consistent with changed functional response only after sensor drift, boundary forcing, pressure effects, and model discrepancy have been excluded.

Rain events therefore convert a long observational record into a sequence of quasi-controlled experiments.

Step I.8: Geometric Phase as an Integral Characterization of a Rain–Dry Trajectory

After the complex state in (12) has been shown to be robust, a Pancharatnam-type phase can be computed for a sequence of time windows. For a closed rain–dry cycle,

$$\gamma_{\text{cycle}} = \arg \left[\prod_{n=1}^M \langle \Psi_n | \Psi_{n+1} \rangle \right], \quad \Psi_{M+1} = \Psi_1. \quad (30)$$

This construction follows the standard logic of geometric phase [19, 20, 21].

The physical interpretation is especially transparent in this context. If the system were instantaneous and memoryless, the same water content would correspond to the same state on the wetting and drying branches:

$$\Psi_{\text{wet}}(\theta_w) = \Psi_{\text{dry}}(\theta_w). \quad (31)$$

The trajectory would essentially retrace itself. In the real system, diffusion, water redistribution, biological response, and reaction create memory:

$$\Psi_{\text{wet}}(\theta_w) \neq \Psi_{\text{dry}}(\theta_w). \quad (32)$$

Accordingly, γ_{cycle} is an oriented descriptor of the normalized response loop, not automatically a “topological invariant,” proof of irreversibility, or identification of a memory mechanism. The measured closure error $d_{\text{FS}}(\Psi_M, \Psi_1)$, minimum adjacent-state overlap, forcing–response coherence, and absolute signal amplitude must be reported with it. If closure error is appreciable, the final factor in (30) is an imposed Pancharatnam/geodesic closure, and the result must not be described as the phase of a physically closed cycle. Because normalization removes common amplitude, every geometric result must also be paired with absolute concentrations or fluxes.

A natural companion measure is the Fubini–Study distance

$$d_{\text{FS}}(\Psi_a, \Psi_b) = \arccos |\langle \Psi_a | \Psi_b \rangle|, \quad (33)$$

which quantifies the difference between two normalized states independently of their overall magnitude.

Step I.9: From a Single Column to a Spatial Map of LEO Once the single-column inversion has been validated, the procedure is repeated for all vertical columns. For each spatial location m , one obtains

$$D_m(\omega, t), \quad \kappa_m(\omega, t), \quad \gamma_m, \quad d_{\text{FS},m}. \quad (34)$$

One can then construct maps of

$$D(x, y, t, \omega), \quad \kappa(x, y, t, \omega), \quad \gamma_g(x, y), \quad (35)$$

which may reveal regions characterized by:

- enhanced transport delay;
- strong wet–dry hysteresis;
- altered diffusion connectivity;
- unusual source/sink balance.

Such spatial products constitute meaningful scientific results even in the complete absence of a topological defect. They transform the analysis from a collection of disconnected sensor curves into a map of dynamical organization.

Step I.10: Comparison of the Three Replicate Hillslopes For the East, Center, and West hillslopes, one can construct comparable state trajectories

$$\Gamma_E, \quad \Gamma_C, \quad \Gamma_W$$

and their corresponding geometric phases

$$\gamma_E, \quad \gamma_C, \quad \gamma_W.$$

In addition to the phase difference

$$\Delta\gamma_{EC} = \text{wrap}(\gamma_E - \gamma_C), \quad (36)$$

one should use a projective distance between states.

At early stages, the trajectories may be expected to be relatively similar because the hillslopes were designed as replicate systems [1]. If their histories diverge, this may appear as a change in inter-hillslope state distance. The three hillslopes provide valuable but limited system-level replication: sensors and repeated events are nested within hillslopes and are not independent landscape replicates. Inference should use a hierarchical event-within-hillslope model, report each hillslope trajectory and uncertainty, and avoid broad population claims from $n = 3$.

Step I.11: Interannual Experiment The most convincing test of the central idea is to compare a series of similar rain events across different years:

$$\Gamma_{2016}, \Gamma_{2017}, \dots, \Gamma_{2026}. \quad (37)$$

For each event, retain

$$\{D_{\text{app}}(\tau), \kappa_{\text{app}}(\tau), S_{\text{res}}(\tau), \gamma_{\text{cycle}}, d_{\text{FS}}(\tau)\}. \quad (38)$$

A natural null hypothesis is that, after conditioning on season and hydrological state, the transport–reaction dynamics are stationary and exhibit no systematic interannual drift. The alternative hypothesis is that the evolving basalt–soil system changes the operator $\mathcal{L}(t)$, producing reproducible drift in physical and geometric observables.

It is essential to compare hydrologically comparable states rather than arbitrary calendar dates. Otherwise, a seasonal loop could be misinterpreted as long-term landscape evolution.

Step I.12: Criterion for Completion of Phase I Phase I should be considered successful if the following chain can be reproduced across several independent rain events and several vertical columns:

$$\text{rain} \rightarrow \text{vertical amplitude/phase response} \rightarrow \{D, \kappa\} \rightarrow |\Psi\rangle \rightarrow \{\gamma_{\text{cycle}}, d_{\text{FS}}\}. \quad (39)$$

Three additional conditions should be satisfied:

1. D and κ remain stable under reasonable changes in the analysis window and frequency band;
2. geometric observables are robust to removal of one sensor and to moderate changes in preprocessing;
3. inter-event differences correlate with independent hydrological variables and cannot be explained solely by sensor artifacts.

This alone would be a substantial result: it would demonstrate that the complex amplitude–phase structure of the CO_2 field contains information about changing transport–reaction organization that is not reducible to mean concentration.

What Is Deliberately Excluded from Phase I Phase I should *not* begin with a search for $W \neq 0$, exceptional points, or non-Hermitian spectral winding. Those questions become justified only after the complex response field has demonstrated reproducibility.

The hierarchy of evidence should be:

$$\text{local phase lag} \rightarrow \text{physical inversion} \rightarrow \text{global geometric trajectory} \rightarrow \text{spatial phase field} \rightarrow \text{topology}. \quad (40)$$

For this reason, the project-specific physical amplitude–phase framework serves as Phase I of the overall project. It must be validated independently of Buchstaber’s results. Successful completion activates, but does not predetermine the conclusions of, Phase II.

6 Phase II of the Project: Integrated Soil-State Modeling, Mathematical Structure, and AI

Objective of Phase II. Phase II asks whether the trustworthy event responses established in Phase I contain reproducible information about longer-term functional reorganization, whether that information covaries with independently measured soil-forming processes, and whether it improves prediction of wholly held-out events. It also has a diagnostic design objective: identify what the present experiment cannot resolve and why.

Phase II work program.

1. connect event-scale apparent transport–exchange and path descriptors to the slow soil-state vector rather than interpreting CO₂ alone;
2. use Buchstaber-style centered unfolding, recurrence, and Grassmann geometry to compare empirical response structure across matched events, years, and hillslopes;
3. test competing physical and statistical explanations with explicit uncertainty, null models, and model-discrepancy diagnostics;
4. train uncertainty-aware AI emulators only for declared targets and validate them on complete held-out events and years;
5. produce an *observability and information-gap map*: which mechanisms, locations, depths, state regimes, or time scales remain confounded under the current forcing, sensor layout, and sampling schedule.

Criterion for completion of Phase II. Phase II is complete when the project can distinguish robust findings from unresolved questions and attach each unresolved question to a concrete experimental cause—for example, missing depth support, inadequate boundary measurements, event mismatch, sparse aqueous chemistry, absent biological sampling, or parameter non-identifiability. Its output is not merely a trained model. It is a validated response atlas, a set of competing hypotheses with uncertainty, and a prioritized evidence base for Phase III experimental-design learning.

7 Why Experiment, Physics, Mathematical Structure, and AI Should Be Combined

No single layer can deliver the project outcome. Controlled experiments provide interventions and validation; physical balances restrict admissible explanations; mathematical geometry creates comparable descriptors across scales; and AI learns multivariable response maps and helps select the next informative test.

Physical and experimental layer. This layer asks: *Which candidate mechanisms are consistent with conservation, boundary conditions, direct samples, and the response to a declared perturbation?* It anchors every derived quantity to forcing, sensor support, uncertainty, and independent observations.

Lag/low-rank analysis. This analysis asks: *How many empirical directions are required to reproduce the measured response at a stated scale and tolerance, and does that representation change after forcing?* Mathematically, the construction is closely related to Singular Spectrum Analysis (SSA), in which a time series is embedded in a Hankel/trajectory matrix, decomposed by SVD, and reconstructed in a low-rank representation [16, 17]. The result is a description of observed response complexity, not a count of physical processes.

Deymier-style geometric-phase analysis. This analysis asks a different question: *How does the orientation of a normalized complex multicomponent state change as environmental parameters vary or as the system traverses a closed cycle?* In acoustic sensing, geometric phase has been used as a global characteristic of changes in the medium; in particular, Deymier and coauthors proposed it for sensing ground-stiffness changes associated with warming permafrost [18]. For LEO, this idea must be transferred nonliterally: the acoustic eigenstate is replaced by a complex state extracted from the spatiotemporal CO₂ field.

AI learning and experimental-design layer. This layer asks: *Can the response to a proposed forcing be predicted with calibrated uncertainty, when should the model abstain, and which next event or sample would best discriminate the remaining hypotheses?* Inputs include declared forcing, antecedent hydrological and thermal state, sensor-quality information, and independently measured slow-state variables. Targets must be explicit—for example, a full event-response curve, relaxation time, or selected response descriptors. Evaluation must distinguish within-regime interpolation from held-out-event, cross-year, and cross-hillslope transfer and must compare against persistence, climatology, and conventional regression baselines.

The analytical sequence is therefore not a one-way cascade but a feedback system:

$$\begin{aligned}
\{\text{forcing, antecedent state, direct samples}\} &\longrightarrow \{C_j(t), \theta_w, T, \text{fluxes}\} \\
&\longrightarrow \{\text{physical fit, unfolding geometry, path descriptors}\} \\
&\longrightarrow \{\text{uncertainty, discrepancy, candidate hypotheses}\} \\
&\longrightarrow \text{AI emulator and active-design ranking} \\
&\longrightarrow \text{prospective perturbation or targeted sampling.}
\end{aligned} \tag{41}$$

The physical, mathematical, and AI branches must challenge and validate one another rather than compete as alternative “universal indices.” A predictive AI model may expose stable relationships or experimental opportunities, but it does not acquire mechanistic authority until those relationships survive physical and prospective experimental tests.

8 Physical Model of a Vertical Column

Linear Harmonic Formulation For a single vertical column, let $z = 0$ denote the surface and $z = L$ the base. If horizontal gradients, variable gas storage, and advection are negligible on the selected interval, consider

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial z^2} - \kappa C + S(z, t), \tag{42}$$

with, for example, an atmospheric Robin condition and a stated basal condition,

$$-DC_z(0, t) = h_g[C(0, t) - C_{\text{atm}}(t)], \quad -DC_z(L, t) = 0, \quad C(z, 0) = C_{\text{init}}(z). \tag{43}$$

Actual LEO boundary conditions must replace these examples. Without boundary and initial conditions the forward and inverse problems are not well posed.

For a coherent harmonic component relative to a known forcing, the homogeneous finite-domain solution is

$$\hat{C}_h(z, \omega) = Ae^{-qz} + Be^{qz}, \quad q^2 = \frac{\kappa + i\omega}{D}. \tag{44}$$

The single decaying exponential used below additionally requires $B \approx 0$ and negligible distributed harmonic source $\hat{S}(z, \omega)$. Otherwise a particular solution and both boundaries must be fitted. Let

$$q = \alpha + i\beta. \tag{45}$$

Then

$$\alpha^2 - \beta^2 = \frac{\kappa}{D}, \quad 2\alpha\beta = \frac{\omega}{D}. \tag{46}$$

Hence

$$D = \frac{\omega}{2\alpha\beta}, \quad \kappa = D(\alpha^2 - \beta^2). \tag{47}$$

Writing

$$\hat{C}(z, \omega) = A(z, \omega)e^{i\phi(z, \omega)}, \tag{48}$$

one obtains the local estimates

$$\alpha \simeq -\frac{\Delta \ln A}{\Delta z}, \quad \beta \simeq -\frac{\Delta \phi}{\Delta z}. \tag{49}$$

Amplitude attenuation and phase lag provide two observables that may constrain transport and exchange under the stated assumptions; they do not uniquely separate them in the presence of distributed sources or phase exchange.

Nearly Pure Diffusion Limit If $\kappa \approx 0$, then

$$q = \sqrt{\frac{i\omega}{D}} = (1+i)\sqrt{\frac{\omega}{2D}}, \quad (50)$$

so that $\alpha = \beta = \sqrt{\omega/(2D)}$ and

$$\phi(z, \omega) = -z\sqrt{\frac{\omega}{2D}}. \quad (51)$$

It follows that

$$D = \frac{\omega}{2(\partial_z \phi)^2}. \quad (52)$$

This gives a first-pass D_{app} only for a coherent, boundary-forced, nearly pure-diffusion component; it is not a general diffusivity estimator.

Physical Interpretation of the Phase Profile Phase becomes a transport delay only after choosing an explicit reference forcing and demonstrating high cross-spectral coherence and non-negligible amplitude. The relevant transfer estimate is, for example, $H_j(\omega) = S_{C_j F}(\omega)/S_{FF}(\omega)$, with $\phi_j = \arg H_j$. A sensor's standalone Fourier phase depends on time origin and on all coherent internal and boundary sources. Under admissible diffusive transport, phase lag should increase systematically with distance; otherwise it is simply a response phase.

In practical terms:

- an increase in source strength may strongly alter amplitude while producing a relatively weak change in phase gradient;
- a decrease in D increases transport delay and $|\Delta\phi|$;
- an increase in κ modifies both attenuation and the complex propagation constant.

These tendencies can reduce equifinality but are not unique fingerprints. Joint amplitude–phase analysis must be combined with moisture, pressure, temperature, gas-filled porosity, aqueous chemistry, and source information.

9 Complex Multichannel State

State in Sensor Space For N sensors, define within each sliding window

$$\hat{C}_j(\omega, t) = A_j(\omega, t)e^{i\phi_j(\omega, t)}. \quad (53)$$

The normalized state is

$$|\Psi(\omega, t)\rangle = \frac{1}{\mathcal{N}} \begin{pmatrix} \hat{C}_1 \\ \hat{C}_2 \\ \vdots \\ \hat{C}_N \end{pmatrix}, \quad \mathcal{N}^2 = \sum_{j=1}^N |\hat{C}_j|^2. \quad (54)$$

Normalization removes the common scale and shifts the analysis from absolute concentration to relative amplitude–phase organization. States differing only by a common complex phase are physically equivalent, so the natural geometry is projective. For normalized pure states, the standard Fubini–Study distance is [22]

$$d_{\text{FS}}(\Psi_a, \Psi_b) = \arccos |\langle \Psi_a | \Psi_b \rangle|. \quad (55)$$

A Vertical Sensor Triplet For a vertical column containing three signals C_1, C_2, C_3 ,

$$|\psi_m\rangle = \frac{(\hat{C}_{m1}, \hat{C}_{m2}, \hat{C}_{m3})^\top}{\sqrt{|\hat{C}_{m1}|^2 + |\hat{C}_{m2}|^2 + |\hat{C}_{m3}|^2}}. \quad (56)$$

Each column is then represented by a point in $\mathbb{CP}^2$. This provides a useful intermediate level between simple pairwise phase lags and a global state for the entire hillslope.

10 Geometric Phase and Hysteresis

Discrete Pancharatnam Construction For a sequence of normalized states $|\Psi_1\rangle, \dots, |\Psi_M\rangle$, it is convenient to use a gauge-invariant phase based on overlaps between neighboring states. Pancharatnam introduced relative phase through the argument of an inner product, and subsequent developments placed this construction within the broader geometric-phase framework [19, 20, 21]. For a closed discrete trajectory,

$$\gamma_{\text{cycle}} = \arg \left[\prod_{n=1}^M \langle \Psi_n | \Psi_{n+1} \rangle \right], \quad \Psi_{M+1} \equiv \Psi_1. \quad (57)$$

For noisy experimental data, this discrete construction is preferable to direct numerical differentiation of $i \int \langle \Psi | \dot{\Psi} \rangle dt$.

The Rain–Dry Cycle as a Natural Closed Path Controlled rainfall manipulation is one of the defining strengths of LEO [1]. A rain event should therefore be treated not as a nuisance but as a controlled perturbation:

$$\text{rain} \rightarrow \theta_w(t) \rightarrow \{D_{\text{app}}(t), \kappa_{\text{app}}(t), S_{\text{res}}(t)\} \rightarrow C_j(t) \rightarrow |\Psi(t)\rangle. \quad (58)$$

If the system were instantaneous and memoryless, states at the same θ_w on the wetting and drying branches would coincide:

$$\Psi_{\text{wetting}}(\theta_w) = \Psi_{\text{drying}}(\theta_w). \quad (59)$$

The path would approximately retrace itself and the closed geometric phase would be small.

The actual LEO system has several sources of memory:

- diffusion times;
- water redistribution;
- microbial response;
- geochemical reaction;
- saturation hysteresis and evolving pore connectivity.

Thus,

$$\Psi_{\text{wetting}}(\theta_w) \neq \Psi_{\text{drying}}(\theta_w), \quad (60)$$

and γ_{cycle} becomes an integral measure of the oriented geometry of the hysteresis loop.

11 Buchstaber-Style Lag/Low-Rank Layer

Lag Embedding and the Trajectory Matrix In this document, “Buchstaber-style” analysis refers to the geometric time-series framework based on multidimensional unfolding, the rank and approximate rank of the unfolding curve, optimization over Grassmann manifolds, and reconstruction of a time series from projections of its unfolding [13, 15]. The numerical use of trajectory matrices, singular values, reconstruction error, and effective rank is closely related to SSA/Hankel methods [16, 17], but the Grassmannian interpretation used below follows Buchstaber’s formulation.

For a regularly sampled time series $x_1, \dots, x_N$ and lag length $L < N$, form the node-oriented trajectory matrix

$$H = \begin{pmatrix} x_1 & x_2 & \cdots & x_K \\ x_2 & x_3 & \cdots & x_{K+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_L & x_{L+1} & \cdots & x_N \end{pmatrix}, \quad K = N - L + 1, \quad (61)$$

whose columns are unfolding nodes. Buchstaber’s affine-rank/scattering construction uses node centering

$$H_c = H - \bar{X} \mathbf{1}_K^T, \quad \bar{X} = K^{-1} H \mathbf{1}_K, \quad (62)$$

not merely subtraction of the scalar-series mean; edge effects make those operations different in finite windows. Raw Hankel rank, SSA reconstruction rank, and Buchstaber affine rank must be reported as distinct objects when they differ. Let

$$H_c = U \Sigma V^*, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq 0. \quad (63)$$

Define the energy fraction captured by the first r modes as

$$V_r = \frac{\sum_{k=1}^r \sigma_k^2}{\sum_k \sigma_k^2}. \quad (64)$$

Then $V_1 \approx 1$ indicates nearly rank-one dynamics, whereas a decrease in V_1 indicates that several independent temporal structures are required to represent the window.

To distinguish clearly between the squared relative error and the relative Frobenius norm, define both:

$$E_r = \frac{\|H - H_r\|_F^2}{\|H\|_F^2} = 1 - V_r, \quad R_r = \frac{\|H - H_r\|_F}{\|H\|_F} = \sqrt{1 - V_r}. \quad (65)$$

Effective Rank Let

$$p_k = \frac{\sigma_k^2}{\sum_j \sigma_j^2}, \quad S_{\text{eig}} = - \sum_k p_k \ln p_k. \quad (66)$$

The effective rank is

$$r_{\text{eff}} = \exp(S_{\text{eig}}). \quad (67)$$

It is an entropy summary of how evenly the selected unfolding energy is distributed among empirical directions. It is not a count of physical processes. A change after rain may arise from changed forcing, nonstationarity, noise, preprocessing, transport, biology, or chemistry; mechanism requires independent evidence.

12 Direct Methodological Consequences of Buchstaber’s Unfolding Theory for the LEO Project

The two Buchstaber papers provide considerably more than a generic justification for using a Hankel matrix. They suggest a concrete hierarchy of objects that can be carried directly into

the LEO analysis. The central change is conceptual: the primary object is not the scalar record $f = (f_1, \dots, f_N)$, but its n -dimensional unfolding

$$X_f = \{X_1, \dots, X_P\} \subset \mathbb{R}^n, \quad X_q = (f_q, f_{q+1}, \dots, f_{q+n-1})^\top, \quad P = N - n + 1. \quad (68)$$

Buchstaber treats X_f as a piecewise-linear curve and studies its geometry through low-dimensional affine subspaces and Grassmann manifolds [13, 15]. This is exactly the representation needed for LEO, because slow evolution, seasonal cycles, rain-driven departures, and structural changes can all be expressed as geometric changes of the unfolding rather than as isolated changes of mean or variance.

Buchstaber rank, approximate rank, and the component-analysis q -width. Let $X_j \in \mathbb{R}^n$, $j = 1, \dots, P$, be the unfolding nodes, $\bar{X} = P^{-1} \sum_j X_j$, and define explicitly the unnormalised scattering matrix

$$W_f = \sum_{j=1}^P (X_j - \bar{X})(X_j - \bar{X})^\top. \quad (69)$$

Let $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ be its eigenvalues. In Buchstaber's formulation the affine rank does not exceed r exactly when $\lambda_i = 0$ for all $i > r$, and the stated ε -rank criterion is

$$\sum_{i=r+1}^n \lambda_i < \varepsilon. \quad (70)$$

Here ε has the squared units of the observations and depends on whether the scattering matrix is divided by P ; this convention must therefore be reported.

For the component-analysis criterion used by Buchstaber, define the empirical summed-squared optimal residual

$$\mathcal{W}_q^{(2)}(X_f) := \min_{\dim L=q} \sum_{j=1}^P \text{dist}(X_j, \bar{X} + L)^2 = \sum_{i=q+1}^r \lambda_i, \quad q < r. \quad (71)$$

This is the quantity called the q -width in the component-analysis result of [15]. It is an optimal *energy* criterion on the unfolding nodes; it must not be conflated with the classical worst-case Kolmogorov width in (76). A scale-preserving version is the RMS residual $d_q^{\text{RMS}} = \sqrt{\mathcal{W}_q^{(2)}/P}$.

For comparison among windows with different total variance, the present project additionally defines

$$\eta_q(t; n) = \frac{\mathcal{W}_q^{(2)}(t; n)}{\sum_{i=1}^n \lambda_i(t; n)}, \quad 0 \leq \eta_q \leq 1. \quad (72)$$

This project-specific normalization separates spectral shape from a common multiplicative change in variance, but it does not remove dependence on sampling interval, embedding length, window, preprocessing, coloured noise, or forcing complexity.

Instead of selecting a model dimension from a single explained-variance threshold, the project will retain the full width profile

$$\mathbf{W}(t; n) = (\mathcal{W}_0, \mathcal{W}_1, \dots, \mathcal{W}_{n-1}) \quad (73)$$

and its normalized counterpart

$$\boldsymbol{\eta}(t; n) = (\eta_0, \eta_1, \dots, \eta_{n-1}). \quad (74)$$

The discrete decrement

$$\mathcal{W}_{q-1} - \mathcal{W}_q = \lambda_q \quad (75)$$

shows exactly how much additional structure is captured by permitting one more dynamical dimension. A pronounced change in the shape of the width profile after rainfall is therefore more informative than a change in a single effective-rank number.

13 Kolmogorov Width as a Central Model-Selection and Change-Detection Principle

The 2025 paper makes a point that is particularly important for the present project: the Kolmogorov-width idea is not simply another way to report the spectrum of a covariance or Hankel matrix. Its meaning is *optimal approximation by a subspace of prescribed dimension*. In the classical formulation, for a set C in a normed space X ,

$$d_q(C; X) = \inf_{L_q} \sup_{x \in C} \inf_{y \in L_q} \|x - y\|_X, \quad (76)$$

where the infimum is taken over q -dimensional affine subspaces. Buchstaber transfers this principle from classes of functions to multidimensional unfoldings of time series and then, for the component-analysis functional, obtains the computable tail-eigenvalue expression (71) [15].

This distinction is fundamental for LEO. A Fourier spectrum asks which frequencies are present. A conventional PCA asks how much variance is carried by successive principal components. The width asks a more direct modeling question:

What is the smallest unavoidable error if the observed dynamics are required to live in only q dimensions?

That question is useful before constructing an empirical reduced representation, but it does not by itself license a low-order physical model.

Width profile, PCA spectrum, and subspace orientation. For the squared-energy criterion, $\eta_q = 1 - V_q$: it is exactly the residual-variance curve obtained from PCA, not additional information beyond the eigenvalue spectrum. Buchstaber's value is the explicit unfolding geometry, affine-rank formulation, reconstruction, and Grassmannian extremal viewpoint. Retaining the full curve avoids hiding the result behind a single arbitrary threshold. In LEO, the combined eigenvalue and eigenvector analysis distinguishes three situations to be separated:

1. the total CO₂ variability increases but the normalized width profile η_q is unchanged: the dynamics have become stronger but not structurally more complex;
2. η_q increases at fixed q : the same q -dimensional model explains a smaller fraction of the unfolding, indicating the activation of additional dynamical structure;
3. η_q remains small but the optimal subspace L_q^* rotates strongly on $G(n, q)$: the effective dimension is unchanged, but the dynamical organization has changed.

The third case is particularly important because it cannot be diagnosed from eigenvalues alone.

An empirical representation dimension from the width curve. For every stated sampling interval, preprocessing choice, window, and embedding length n , define

$$q_*(t; n) = \min \{q : \eta_q(t; n) \leq \varepsilon_{\text{str}}\}, \quad (77)$$

where ε_{str} is calibrated from sensor noise and surrogate or held-out data. Stability under moderate changes of n is required. The correct operational chain is

$$\begin{aligned} \text{width profile} &\longrightarrow \text{empirical response dimension at a stated scale} \\ &\longrightarrow \text{candidate reduced representation and screening diagnostic.} \end{aligned} \quad (78)$$

It does *not* determine the differential order, state dimension, or number of physical processes in the underlying reaction-transport system. A high-dimensional PDE under simple forcing can produce low-rank output; a simple nonlinear or nonstationary system can produce high-rank output.

What the component-analysis width actually bounds. If $\mathcal{W}_q^{(2)}$ is large, no q -dimensional affine subspace can represent the chosen unfolding nodes with small summed-squared error. This is a rigorous statement about those observations under that criterion, not about every q -parameter physical explanation. Buchstaber’s result is nevertheless valuable before unstable nonlinear approximation because it quantifies the best empirical compression attainable at fixed q [15]. For LEO it can flag response windows that differ from a baseline and guide reduced-order regularization. Physical-model adequacy must be tested separately using conservation residuals, explicit boundary conditions, parameter identifiability, out-of-sample prediction, and independent hydrological and geochemical measurements.

Width across controlled rain events. Buchstaber’s formulation also allows a q -width to be associated with a *set* of time series by considering the union of their unfolding curves. This suggests a particularly strong LEO experiment. Let $\mathcal{E}$ be a collection of hydrologically matched rain events and let

$$\mathcal{X}_{\mathcal{E}} = \bigcup_{e \in \mathcal{E}} X_{f_e}. \quad (79)$$

Then one can ask whether a common q -dimensional affine subspace represents all events with small error. A small event-family width means that the same low-dimensional organization is shared across repeated perturbations. A growing width after several years means that no single low-dimensional subspace can represent all responses as well as before.

For the project, two versions should be retained:

1. a worst-case Kolmogorov-type width based on the maximum distance of an unfolding node from the candidate subspace;
2. an empirical energy width based on the summed squared distances and therefore directly compatible with the eigenvalue formula (71).

The first is sensitive to a localized but reproducible structural excursion; the second is more stable in the presence of measurement noise. Agreement between them would provide strong evidence that a detected change is not an artifact of the chosen loss function.

Width difference as a quantitative response to forcing. For event e , define pre-rain and post-rain normalized profiles

$$\eta_q^{\text{pre}}(e), \quad \eta_q^{\text{post}}(e). \quad (80)$$

The project-specific structural response is

$$\Delta\eta_q(e) = \eta_q^{\text{post}}(e) - \eta_q^{\text{pre}}(e). \quad (81)$$

A transient positive $\Delta\eta_q$ followed by recovery indicates a temporary increase in the empirical representation error at the selected q . A persistent positive shift means that, under the same embedding and tolerance, the measured post-event response is less well represented by the previous affine description; it does not count physical degrees of freedom. The recovery time

$$\tau_{\mathcal{W}} = \inf \{ \tau > 0 : |\eta_q(t_{\text{rain}} + \tau) - \eta_q^{\text{pre}}| < \delta_{\mathcal{W}} \} \quad (82)$$

provides a structural relaxation time that can be compared directly with moisture relaxation, the phase-lag-derived diffusion time, and changes in κ .

A new use of the Grassmann optimization algorithm in the 2025 paper. Buchstaber does not restrict the Grassmannian framework to PCA. He formulates general criteria as functions on $G(n, q)$ and gives a gradient-optimization construction based on elementary orthogonal rotations $A(t; k, l)$ that generate local coordinate directions on the Grassmann manifold [15]. This opens a substantial methodological extension for LEO.

The first pass should use the PCA-optimal subspace L_q^* . In a second pass, the project can define a *physically informed* Grassmann objective

$$\mathcal{J}(L) = w_1 \frac{\|r_f(L)\|^2}{\|f - \bar{f}\|^2} + w_2 \mathcal{E}_{\text{RD}}(L) + w_3 \mathcal{E}_\phi(L), \quad L \in G(n, q), \quad (83)$$

where $\mathcal{E}_{\text{RD}}$ measures inconsistency of the reaction–diffusion estimates across adjacent depth pairs and $\mathcal{E}_\phi$ measures instability of the extracted phase under neighboring windows or sensor omission. The weights w_j must be selected prospectively or through cross-validation.

Equation (83) is *not* proposed in Buchstaber’s paper; it is a project-specific extension made possible by his observation that approximation, classification, and other criteria can be optimized directly on $G(n, q)$. Its advantage is that the selected subspace need not be merely the one that retains the most variance. It can be the low-dimensional subspace that gives the most reproducible *physical* description of the LEO response.

Kolmogorov width versus reconstruction error. The two quantities should not be conflated. A reconstruction error such as

$$R_q = \frac{\|H - H_q\|_F}{\|H\|_F} \quad (84)$$

is tied to a particular matrix representation and normalization. The width is formulated as an optimization over all admissible q -dimensional subspaces of the unfolding geometry. With the Euclidean/PCA functional the two are closely related computationally, but the width language makes explicit the deeper statement: *no alternative q -dimensional affine description can do better under the chosen criterion*. This gives the quantity a direct model-selection interpretation rather than only a compression interpretation.

Kolmogorov width versus effective rank. Effective rank compresses the complete spectrum into one entropy-like scalar. That is useful for visualization, but it discards information about *where* dimensional complexity enters. The sequence

$$\mathcal{W}_0, \mathcal{W}_1, \dots, \mathcal{W}_{n-1} \quad (85)$$

retains the full hierarchy of approximation errors. Consequently, two windows can have similar effective rank but substantially different width profiles. For the project, effective rank should therefore remain a summary variable, while the width profile should be the primary model-order diagnostic.

Kolmogorov width and physical interpretability. The width itself is not a transport coefficient and must not be interpreted as one. Its role is structural. The physical meaning appears when width changes are compared with independently estimated transport–reaction quantities:

$$\Delta\eta_q \leftrightarrow \Delta D_{\text{app}}, \Delta\kappa_{\text{app}}, \Delta S_{\text{res}}, \Delta\gamma_g. \quad (86)$$

This separation is an advantage: the width first states, without committing to a mechanism, that the affine representation error or required empirical dimension has changed. The restricted physical analysis then tests which candidate explanations remain admissible.

Specific hypothesis enabled by the width concept. A new, directly testable hypothesis can therefore be added:

As granular basalt evolves toward a more heterogeneous biologically and geochemically active medium, hydrologically matched CO_2 responses will require a progressively richer low-dimensional representation. This will appear as a systematic upward shift of normalized q -widths at fixed small q , an increase in q_* , or a persistent rotation of the optimal Grassmann subspace even when mean CO_2 concentration changes only weakly.

The hypothesis can fail. If the width profiles and optimal subspaces remain stable across hydrologically matched years, then the proposed structural evidence for long-term reorganization is absent.

Why this materially strengthens the project. The Kolmogorov-width layer gives the project three properties that are otherwise difficult to obtain simultaneously:

1. **Optimality:** it quantifies the best attainable approximation at a prescribed dynamical dimension rather than the performance of one arbitrarily selected basis.
2. **Model-order control:** it supplies a data-driven bound on how simple a physical model is allowed to be before nonlinear parameter estimation begins.
3. **Comparability:** normalized width profiles can be compared among windows, rain events, sensor columns, hillslopes, and years while retaining the complete hierarchy of unresolved dimensions.

For LEO, this makes the width concept more than an auxiliary statistic. It becomes the structural gatekeeper between raw time-series observations and the physical/geometric interpretation that follows.

The optimal subspace as a point on a Grassmann manifold. For each time window and each selected q , let

$$L_q^*(t) \in G(n, q) \quad (87)$$

be the q -dimensional subspace spanned by the leading principal directions of the unfolding. Buchstaber formulates the dimension-reduction problem as an extremal problem on $G(n, q)$: the optimal L_q^* maximizes the statistical spread retained by projection [15]. This observation gives the LEO project an additional state variable that is absent from a singular-value-only analysis: the *orientation* of the optimal subspace.

Accordingly, two windows can have nearly identical spectra but different leading subspaces. Such rotation is an empirical change that may have physical causes, but those causes are not identified by geometry alone. Moreover, L_q^* is unique only when $\lambda_q > \lambda_{q+1}$; a small eigengap makes its orientation noise-sensitive. Grassmann comparisons therefore require a reproducible eigengap, bootstrap uncertainty for the projector $P_q = U_q U_q^T$, and consistent centering and embedding conventions.

A practical Grassmannian distance can therefore be introduced from the principal angles $\theta_1, \dots, \theta_q$ between two optimal subspaces:

$$d_G(L_q^*(t_1), L_q^*(t_2)) = \left(\sum_{j=1}^q \theta_j^2 \right)^{1/2}. \quad (88)$$

Equation (88) is a project-level diagnostic motivated by Buchstaber's Grassmannian formulation; it is not claimed to be a new definition of distance on a Grassmann manifold. Its role is to quantify rotation of the dominant dynamical subspace.

A rank-preserving recurrence as a local diagnostic model. For an *exact* affine-rank result with $r \leq n - 1$, Buchstaber's theory yields a finite affine recurrence on the index range specified by the theorem. If the unfolding rank does not exceed r , then there exist constants $c_0, c_1, \dots, c_k$, with $k \leq r$, such that

$$f_m = \sum_{s=1}^k c_{k-s+1} f_{m-s} + c_0. \quad (89)$$

The recurrence follows from the Schubert-basis description of the orthogonal complement to the supporting affine plane [13, 15]. The exact rank-preserving-extension theorem is an if-and-only-if statement only when one recurrence holds over the required whole observed range. LEO data are noisy and only approximately low rank, so forecast failure is a project-level change-point heuristic, not a conclusion of the theorem and not physical proof.

For the LEO project, equation (89) should therefore be fitted with regularization, coefficient uncertainty, held-out forecasts, and matched null surrogates, locally rather than used as a global law. In every sliding window:

1. estimate the lowest stable rank r ;
2. estimate the corresponding recurrence coefficients;
3. propagate the recurrence a short distance beyond the fitting interval;
4. compute the one-step and multi-step continuation residuals;
5. compare the residual increase with rain onset, moisture redistribution, temperature, and the physically inferred D and κ .

A sharp failure of rank-preserving continuation is then a mathematically defined candidate for a dynamical regime change. This gives a much more specific structural-change test than a generic threshold on variance or mean concentration.

Finite-rank functions and model-class selection. Buchstaber establishes a conditional, one-way chain: a continuous function satisfying the stated finite separable translation equation has finite rank; with the paper’s differentiability assumptions it satisfies a constant-coefficient ODE, whose solutions are exponential–polynomial–oscillatory [13, 15]. This does not imply that every approximately low-rank environmental record is generated by such an ODE, nor does its rank count mechanisms.

The practical consequence is narrower but useful: empirical rank can constrain how many components are needed in a chosen finite-rank approximation class. It cannot validate the reaction–transport PDE. A rise after rain says that the old empirical representation is less adequate at that scale; forcing, nonstationarity, and noise must be excluded before giving it process meaning.

L -approximation and reconstruction from an arbitrary dynamical subspace. A major addition in the 2025 paper is the systematic construction of an L -approximation. For any q -dimensional subspace $L \subset \mathbb{R}^n$, one first projects the unfolding:

$$X_f \longrightarrow \pi_L X_f, \quad (90)$$

and then projects the resulting curve back onto the linear space of legitimate time-series unfoldings. This produces a reconstructed series

$$\hat{f}_L = \Pi(\pi_L X_f) \quad (91)$$

and a residual

$$r_f(L) = f - \hat{f}_L. \quad (92)$$

Thus $\|r_f(L)\|$ is itself a function on $G(n, q)$ [15].

This construction should be used explicitly in LEO. Three distinct choices of L are useful:

1. $L = L_q^*$, the optimal variance-preserving subspace;
2. $L = L_{\text{phys}}$, a subspace spanned by modes associated with selected physically meaningful time scales, such as diurnal and rain-response components;
3. $L = L_{\text{ref}}$, a reference subspace learned from a baseline dry or early-year regime.

The three residuals answer different empirical questions. The first is the residual from a variance-optimal node subspace; it is not proved to be the irreducible error of a best rank- q time series because projection back to the Hankel/time-series subspace can change rank. The second measures the part not represented by selected time-scale modes, and the third departure from a baseline subspace. All three also contain noise and preprocessing error.

Why the residual is scientifically useful. The 2025 paper demonstrates the L -approximation on a long atmospheric CO₂ record, using $n = 12$ and reconstructions from the first $k = 1, 3, 5$ principal directions. The corresponding residuals separate trend and annual/semiannual cyclic structure from smaller unexplained components [15]. For the LEO problem, the residual should not be treated merely as “noise.” It is a candidate carrier of the processes not represented in the selected low-dimensional state.

For a vertical LEO column, the project should therefore analyze

$$r_C^{(m)}(t; L_q^*), \quad r_C^{(m)}(t; L_{\text{phys}}), \quad r_C^{(m)}(t; L_{\text{ref}}) \quad (93)$$

jointly with moisture, temperature, and the physically reconstructed source term. A residual that grows systematically during wetting but collapses during drying would indicate a process with asymmetric activation that is invisible in a simple low-rank reconstruction.

The Fourier–unfolding connection is a one-way consistency check. Buchstaber’s 1994 and 2025 papers show that a nondegenerately sampled pure sinusoid has affine unfolding rank at most two. Degenerate sampling, including frequencies equivalent to 0 modulo π for a sampled sine, can lower the rank. In ambient Euclidean geometry the locus is generally an ellipse; a regular polygon requires special commensurability and sample conditions [13, 15].

Thus an oscillatory component implies an approximately planar unfolding under suitable sampling, but the converse is false: two exponentials, polynomial/recurrence structure, and other second-order sequences can also have rank two. Approximate planarity is therefore only a necessary consistency check to be combined with spectral coherence, stationarity, aliasing, and surrogate tests.

The defensible bridge is

$$\begin{aligned} \text{coherent nondegenerate sinusoid} &\implies \text{affine rank at most two,} \\ \text{complex/projective phase layer} &= \text{a separate project construction.} \end{aligned} \quad (94)$$

14 Concrete Buchstaber–Deymier Implementation for LEO

The project should be implemented in a nested hierarchy. The hierarchy is important because it prevents a sensitive geometric or topological observable from being calculated on a dynamical component that has not first been shown to be structurally reproducible.

Stage A: establish the unfolding geometry for every sensor and every vertical column. For each reliable signal and for each window length n , construct the unfolding (68). Repeat the analysis over a physically meaningful set of embedding lengths rather than fixing a single n . For a monthly atmospheric CO₂ record, Buchstaber used $n = 12$ to resolve annual structure; in LEO, n should instead be tied to the target physical scale. For example:

- diurnal analysis: choose n spanning approximately one to several days;
- rain response: choose n spanning the rise, redistribution, and early drying phases;
- slow evolution: use longer windows spanning weeks to months.

The result is not one rank but a scale-dependent surface

$$r_\varepsilon(t, n). \quad (95)$$

Stage B: compute the full structural signature. For each (t, n) , compute

$$\mathcal{B}(t, n) = \{ \lambda_i, \mathcal{W}, \boldsymbol{\eta}, q_*, r_\varepsilon, L_q^*, d_G(L_q^*, L_{q,\text{ref}}^*), \|r_f(L_q^*)\| \}. \quad (96)$$

The vector $\mathcal{B}$ is a Buchstaber structural signature. It distinguishes four effects that a single effective rank does not:

1. change in total modal energy;
2. change in effective dimension;
3. rotation of the dominant subspace;
4. increased distance from the best low-dimensional reconstruction.

Stage C: detect structural disruptions by rank-preserving continuation. Estimate the recurrence (89) inside each stable window. Define

$$E_{\text{cont}}(t) = \frac{\sum_{j=1}^h [f_{t+j} - \hat{f}_{t+j}^{(\text{rec})}]^2}{\sum_{j=1}^h [f_{t+j} - \bar{f}]^2 + \eta}, \quad (97)$$

where h is a short continuation horizon and $\eta > 0$ is a small numerical regularizer. A rapid increase in E_{cont} , accompanied by an increase in q -width or by a rotation of L_q^* , is a candidate empirical change that requires robustness and physical follow-up.

For LEO, such events should be aligned with:

$$\{\text{rain onset}, \theta_w, T, P_{\text{CO}_2}, D_{\text{app}}, \kappa_{\text{app}}, S_{\text{res}}\}. \quad (98)$$

This alignment makes the statistical label testable against physical covariates; it does not by itself assign a mechanism.

Stage D: perform physical inversion only on structurally admissible windows. The reaction–diffusion inversion for D and κ should be restricted to windows for which:

$$r_\varepsilon \text{ is stable}, \quad d_q \text{ is small for the chosen } q, \quad E_{\text{cont}} \text{ is not anomalously large}. \quad (99)$$

This is an empirical regularity screen against indiscriminate fitting, not a physical model-admissibility theorem. The PDE assumptions, boundary conditions, coherence, identifiability, and conservation residual must be tested independently.

Stage E: construct the geometric-phase state in the robust subspace. Once L_q^* is identified, compute complex modal coefficients $a_k = A_k e^{i\vartheta_k}$ and form

$$|\Psi_B\rangle = \frac{(a_1, \dots, a_q)^\top}{\sqrt{\sum_{k=1}^q |a_k|^2}}. \quad (100)$$

This complex state is a project-specific construction, not a Buchstaber object. SVD vectors have arbitrary signs, and nearly degenerate singular blocks may rotate arbitrarily between windows. Before comparing a_k , bases must be aligned by orthogonal/unitary Procrustes transport or an equivalent gauge convention; only eigengap-separated modes or invariant spectral blocks may be tracked. Otherwise apparent modal phase can be a basis artifact.

A second, complementary geometric object is the path

$$t \mapsto L_q^*(t) \in G(n, q). \quad (101)$$

The project therefore contains two different geometries:

1. projective geometry of the complex state $|\Psi_B\rangle$;
2. Grassmann geometry of the evolving low-dimensional dynamical subspace L_q^* .

Their agreement or disagreement is itself informative. A large projective phase with little Grassmann rotation suggests evolution mainly inside a fixed modal subspace. A large Grassmannian displacement indicates that the dynamical basis itself has changed.

Stage F: event-to-event and year-to-year comparison. For every controlled rain event e , construct the event descriptor

$$\mathcal{E}_e = \{D_e(\tau), \kappa_e(\tau), S_{\text{bio},e}(\tau), \boldsymbol{\eta}_e(\tau), q_{*,e}(\tau), r_e(\tau), d_G(\tau), E_{\text{cont}}(\tau), \gamma_e, d_{\text{FS},e}\}. \quad (102)$$

Comparison should be made between events with similar initial moisture and comparable forcing. Long-term landscape evolution is then tested as a systematic drift of the descriptor $\mathcal{E}_e$, not merely as a change in average concentration.

Stage G: spatial extension across the three hillslopes. At each sensor column m , construct $\mathcal{B}_m(t, n)$ and $\mathcal{E}_{e,m}$. The resulting maps include

$$d_q(x, y, t), \quad r_e(x, y, t), \quad d_G(x, y, t), \quad E_{\text{cont}}(x, y, t), \quad D(x, y, t), \quad \kappa(x, y, t), \quad \gamma_g(x, y, t). \quad (103)$$

Spatial coincidence between a structural disruption, a transport anomaly, and a geometric phase anomaly would provide substantially stronger evidence than any one diagnostic alone.

15 How the Framework Describes the Transformation of Crushed Basalt into Soil

The scientific target of LEO is not merely the evolution of a gas concentration field. The underlying process is the progressive transformation of an initially comparatively homogeneous, mineral, porous basalt substrate into a chemically altered, hydrologically structured, and biologically colonized soil. The original LEO design explicitly framed the experiment in terms of the coupled evolution of soil physical and chemical properties, biological communities, hydrologic pathways, and carbon exchange [1]. Subsequent work has confirmed several distinct components of that transformation: early mineral dissolution and secondary mineral formation [4], diffusion-limited supply of pore-space CO_2 to weathering reactions [5], depth-dependent microbial colonization [8], precipitation-driven restructuring of microbial communities [10], and measurable hillslope-scale carbon removal by granular basalt weathering [3]. More recent work shows enrichment of weathering-generated Al- and Fe-bearing colloidal material after approximately twelve years, consistent with secondary-phase development [12].

Operational meaning of “soil formation.” No CO_2 -derived metric defines soil. Here gradual basalt-to-soil transformation is independently evidenced by coordinated changes in primary-mineral abundance, secondary phases and fines, organic-carbon stocks, aggregation and depth structure, bulk density and porosity, water retention and hydraulic behaviour, nutrient-retention capacity, biological colonization, and dissolved or particulate export. The proposed mathematics measures the *functional carbon-response dynamics*; it can support a pedogenic interpretation only when those direct indicators change coherently.

A minimal slow state vector is

$$\boldsymbol{\xi}(z, t) = (M_{\text{primary}}, M_{\text{secondary}}, C_{\text{org}}, \phi, \tau, B), \quad (104)$$

containing primary-mineral mass, secondary-phase/fine mass, organic carbon, porosity, tortuosity or connectivity, and a biological state. Its schematic evolution is

$$\partial_t \boldsymbol{\xi} = \mathbf{F}_{\text{slow}}(\boldsymbol{\xi}, \theta_w, T, \text{pH}, C_{\text{DIC}}, \mathbf{q}_w, \dots), \quad (105)$$

where the components include weathering, precipitation, organic inputs and decay, aggregation, erosion, and solute/colloid export. The fast gas observations depend on this slow state only through constitutive and source relations, for example

$$D_g = D_g(\theta_g, \phi, \tau, \xi), \quad J_{gw} = J_{gw}(C_g, C_{\text{DIC}}, \theta_w, T, \xi), \quad S_g = S_g(T, \theta_w, B, C_{\text{org}}). \quad (106)$$

A corresponding aqueous carbon balance is required for mechanism:

$$\frac{\partial(\theta_w C_{\text{DIC}})}{\partial t} = \nabla \cdot (\theta_w D_w \nabla C_{\text{DIC}}) - \nabla \cdot (\mathbf{u}_w C_{\text{DIC}}) + J_{gw} - R_{\text{aq}}, \quad (107)$$

where R_{aq} includes aqueous speciation and mineral reactions. Gas depletion after wetting can therefore represent reversible dissolution or export as well as mineral consumption; net carbon removal cannot be inferred from pore-gas depletion alone.

Interpretation for ecologists and Earth scientists. Repeated rain events act as probes of the evolving porous medium. Attenuation and phase lag summarize how a *coherent, identified* CO_2 fluctuation is transmitted between depths under the reduced model. Unfolding widths summarize how accurately the measured response can be represented by a low-dimensional empirical subspace. Grassmann and Fubini–Study distances compare changes in the relative mixture of lag or sensor responses, and geometric phase summarizes the orientation of a sufficiently closed normalized wetting–drying loop. Together these quantities can reveal when and where the functional response has changed and can guide direct sampling. They cannot, from gas CO_2 alone, identify a mineral pathway, microbial taxon, weathering rate, or prove that basalt has become soil.

The proposed framework therefore interprets the observations as possible changes in the *transport–exchange organization* of the porous medium, to be tested against (104) rather than equated with it. The central idea is that soil formation should alter not only the mean CO_2 concentration, but also the way perturbations propagate through the depth profile. Repeated wetting and drying probe the evolving medium in much the same way that a controlled input probes a dynamical system.

Stage 1: initially crushed basalt. At the beginning of the experiment, the substrate is relatively homogeneous crushed basaltic tephra with low organic carbon and limited biological complexity. In this regime, the dominant controls on pore-gas CO_2 should be gas diffusion, dissolution into pore water, mineral reaction, and externally imposed hydrological redistribution. Earlier LEO measurements showed that, after rainfall, pore-space CO_2 can be depleted rapidly into solution and that the subsequent supply of CO_2 may become diffusion limited [5]. In the present framework, such a regime should appear as a comparatively low-dimensional unfolding with a stable width profile, reproducible phase lag with depth, and a physically interpretable effective diffusivity D_{eff} .

Stage 2: emergence of geochemical heterogeneity. As weathering proceeds, primary minerals dissolve and secondary phases precipitate. Pore-water chemistry from LEO has already shown spatial gradients in mineral transformation along hydrological flow paths [4]. The recent colloid study further demonstrates that weathered LEO basalt contains substantially more Al- and Fe-bearing colloidal material than pristine basalt, consistent with ongoing generation of reactive secondary phases [12]. These changes can modify pore geometry, surface reactivity, aqueous speciation, and the effective sink for CO_2 .

In the proposed analysis, such evolution should appear through changes in

$$D_{\text{eff}}(z, t), \quad \kappa_{\text{eff}}(z, t), \quad \eta_q(t; n), \quad L_q^*(t), \quad (108)$$

where the fitted D_{eff} and κ_{eff} are frequency-, moisture-, and model-dependent apparent transfer parameters. In particular, κ_{eff} combines reversible gas–water storage, aqueous export, and reaction and is not a weathering-rate constant. The quantities η_q and L_q^* describe empirical approximation error and lag-subspace orientation, not mineralogical state.

Stage 3: biological colonization and biogeochemical feedback. Microbial work at LEO and miniLEO has shown that microbial community structure develops systematically with depth and that hydrological perturbations alter community assembly and potential function [8, 10]. The biological contribution therefore cannot be represented simply by adding a constant respiration term. As colonization develops, microbial production and consumption of carbon, organic exudates, mineral dissolution, and secondary-mineral precipitation can feed back on one another.

Biological measurements may constrain a true source $S_{\text{bio}}(z, t)$, while the CO_2 -only inverse problem yields only S_{res} . Biological development may also change transport and exchange indirectly through organic matter, aggregation, water retention, mineral surfaces, and pore structure. Under the restricted coherent boundary-forced model, source and transport changes can have different tendencies. In the full hillslope, however, distributed respiration, gas–water exchange, advection, and changing gas-filled porosity can mimic those tendencies; attribution requires the independent variables in (104)–(107).

Stage 4: an increasingly soil-like dynamical system. The transition from crushed basalt to soil should ultimately be understood as the emergence of coupled heterogeneity: mineralogical, hydrological, chemical, and biological. In this framework, that transition is expressed not by one scalar “soil formation index,” but by a coordinated change in several observables:

$$\{D_{\text{app}}, \kappa_{\text{app}}, S_{\text{res}}, \boldsymbol{\eta}, q_*, L_q^*, \gamma_g, d_{\text{FS}}\}. \quad (109)$$

The expected mathematical signature is a reproducible change in one or more response descriptors under matched forcing; no sign is prescribed. Added heterogeneity may increase empirical dimension or hysteresis, while stronger coupling, aggregation, or a dominant preferential/root pathway may decrease dimension or synchronize the response. These are two-sided hypotheses.

A multivariate descriptor can detect a change even when competing effects cancel in mean concentration, but it does not by itself detect “soil.” The basalt-to-soil interpretation is supported only when descriptor drift covaries with independent change in $\boldsymbol{\xi}$, aqueous/solid mass balances, and biological observations, and when sensor drift, altered forcing, temperature, pressure, and antecedent hydrology are rejected as explanations.

16 Relation to Existing LEO Approaches and the Added Value of the Proposed Project

Existing LEO research has already established several essential pieces of the basalt-to-soil problem. The proposed project should be viewed as complementary to these approaches rather than as a replacement for them.

Hydrological approaches. LEO studies have quantified storage, runoff, flow paths, and water-age or transit-time behavior under controlled rainfall [6, 11]. These methods describe how water moves through the hillslope and are essential for understanding contact time between water and mineral surfaces. Their principal observable, however, is water transport. They do not by themselves resolve how the gas-phase CO_2 response reorganizes as reaction and biological activity develop.

The present project adds a continuously sampled gas-phase dynamical layer. It asks whether the same rainfall forcing produces the same vertical CO_2 propagation pattern, phase delay, and low-dimensional response geometry after years of landscape evolution.

Geochemical and mineralogical approaches. Pore-water chemistry has revealed spatial gradients in dissolution and secondary-mineral formation [4]; controlled laboratory studies using the same basalt have shown that organic carbon can modify weathering and chemical denudation [7]; and recent LEO work has directly identified weathering-generated colloidal phases [12]. These

methods provide chemically specific evidence of mineral transformation and are indispensable for validating mechanism.

Their limitation is primarily temporal and spatial sampling density: aqueous chemistry, solid sampling, and colloid measurements are necessarily much sparser than the continuously recorded sensor network. The proposed project uses high-frequency CO₂ profiles as a continuous dynamic proxy and then validates the inferred transitions against these chemically specific measurements. Its value is therefore not greater chemical specificity, but greater *temporal continuity and dynamical sensitivity*.

Carbon-balance and weathering approaches. LEO studies have demonstrated that basalt weathering can consume atmospheric carbon and that CO₂ supply to pore spaces may limit weathering [5]. The 2025 hillslope-scale mass-balance experiment further quantified carbon removal through bicarbonate export and showed that water saturation and infiltration influence CO₂ uptake and weathering rate [3]. These studies answer the question: *How much carbon is removed, and under what hydrological conditions?*

The proposed project adds a different question:

How does the internal dynamical organization that controls that removal change through time?

Instead of estimating only an integrated carbon-removal flux, it estimates interval- and frequency-dependent apparent response parameters under declared assumptions and tests whether repeated events remain representable by the same empirical response subspace.

Microbial and ecological approaches. Microbial studies have identified depth-dependent community structure, deterministic and stochastic assembly processes, and precipitation-induced responses [8, 10]. These approaches directly characterize the biotic component of incipient soil formation, but sequencing necessarily produces discrete snapshots.

The proposed framework does not infer microbial taxa from CO₂ data. Rather, it supplies continuous descriptors of the subsurface carbon-response system. Changes in S_{res} , width profile, or event hysteresis can then be compared with independent microbial and respiration measurements. Only those independent constraints can support a biological-source interpretation. This creates a bridge between sparse biological observations and continuous physical sensing.

Surface-flux approaches. LEO measurements have shown that negative soil CO₂ fluxes can arise from non-biological processes and that temperature gradients and subsurface concentration inversions strongly affect the observed surface exchange [9]. This is directly relevant to the present project because it demonstrates that surface flux alone cannot be interpreted as respiration.

The proposed depth-resolved approach addresses precisely this ambiguity. By using phase and amplitude relationships among multiple depths, it adds constraints on internal transport delay and exchange that are unavailable from a single surface-flux measurement, while retaining equifinality unless boundary and source terms are independently constrained.

What is genuinely added by the proposed project. The distinctive contribution is the integration of four levels that have generally been treated separately in the existing LEO literature:

$$\begin{aligned} \text{controlled forcing} &\longrightarrow \text{continuous depth-resolved CO}_2 \text{ response} \\ &\longrightarrow \text{empirical lag geometry and apparent transfer parameters.} \end{aligned} \tag{110}$$

The width/Grassmann layer screens empirical compressibility and subspace change; the restricted amplitude-phase inversion estimates model-dependent apparent parameters only in admissible windows; and the geometric-phase layer describes normalized path orientation and hysteresis. The same analysis can be repeated for comparable rain events over many years.

The resulting method is therefore especially suited to the original LEO goal of observing *coevolution*. It does not examine hydrology, geochemistry, microbiology, or carbon exchange in isolation. Instead, it uses the continuous CO₂ response as an integrating observable and tests its interpretation against independent measurements from those established approaches.

17 Tangible Project Outputs, Users, and Success Criteria

The project will be judged by whether it creates usable and validated capabilities, not by the number of mathematical descriptors produced. The principal deliverables are:

1. **A sensor-quality audit and dated failure calendar.** This living product records objective failure criteria and dated intervals of reliable operation, suspected degradation, failure, maintenance, recovery, and replacement for each analyzed channel. *Success criterion:* every analysis mask is traceable to the calendar, and LEO operators receive a prioritized list of channels requiring verification, recalibration, redundancy, or maintenance action.
2. **A quality-controlled LEO perturbation-response catalogue.** This product will synchronize rainfall and atmospheric forcing, depth-resolved CO₂, moisture, temperature, storage and flux observations, sensor calibration and quality metadata, antecedent-state labels, and links to direct samples. *Success criterion:* every retained event has machine-readable provenance and an independent user can reproduce a benchmark event from the documented raw inputs.
3. **A standardized rain-pulse and sampling protocol.** The protocol will define forcing tolerances, antecedent-state matching, sensor checks, exclusion rules, and timing of aqueous, solid, physical, and biological samples. *Success criterion:* a prospectively declared, power- or information-justified set of matched events is executed across feasible hillslopes with forcing mismatch and exclusions reported.
4. **A validated physical–mathematical inference package.** The software will include finite-domain forward models, model-admissibility gates, apparent D and κ , unresolved-balance estimation, centered unfolding geometry, uncertainty propagation, and synthetic benchmarks. *Success criterion:* it meets predeclared recovery and interval-coverage targets in admissible synthetic cases and reliably abstains or rejects inference when assumptions are violated.
5. **A LEO soil-function response atlas and sampling triggers.** The atlas will show event-conditioned transport/exchange response, relaxation, empirical subspace change, and hysteresis with sensor support and uncertainty. It will not display interpolated certainty where no observation model exists. *Success criterion:* trigger rules prospectively identify candidate locations or times for direct sampling at a controlled false-alert rate.
6. **An independent basalt-to-soil validation analysis.** Fast response descriptors will be tested against primary and secondary minerals, fines and colloids, organic carbon, pore structure and water retention, biological colonization, and dissolved/particulate mass balances. *Success criterion:* preregistered models are evaluated against hydrology-, temperature-, and sensor-age null models on held-out events or years; a well-supported null result remains a valid scientific outcome.
7. **An uncertainty-aware AI event emulator and active-design module.** The emulator will predict a declared response target from forcing and antecedent state, quantify aleatoric and epistemic uncertainty, flag out-of-distribution conditions, and rank candidate next experiments or samples by expected information gain. *Success criterion:* it improves on persistence, climatology, and conventional regression in held-out tests, satisfies a predeclared calibration target, abstains under known distribution shift, and at least one recommendation is tested prospectively.

8. **Evidence-based improvement of the LEO experimental design.** As an additional result—distinct from the AI model itself—the project will combine its sensor-failure calendar, observability and information-gap maps, physical discrepancies, and direct-sample needs into ranked priorities for forcing protocols, sensor locations and depths, sentinel validation hubs, trigger-based sampling, calibration controls, and strategic manipulations at three cost/effort tiers. *Success criterion:* investigators, leadership, and operators jointly select at least one feasible improvement and compare it prospectively with a predeclared standard design.
9. **A transferable Earth-science toolkit.** Documented schemas, versioned workflows, tutorials, synthetic data, validation standards, and portability guidance will allow use beyond LEO. *Success criterion:* an unaffiliated user reproduces the benchmark and the workflow is evaluated on at least one external or deliberately shifted data setting, with failures and required observation conditions documented.
10. **An educational and research-training module integrated with MATH 485/585.** Quality-controlled LEO case studies, mathematical derivations, computational notebooks, and reproducibility exercises will connect the University of Arizona’s mathematical-modeling curriculum with active Earth-system research. *Success criterion:* student analyses are independently reproduced and reviewed, and at least one course-generated question, diagnostic, or software contribution enters the formal research pipeline after validation.

These products create four connected forms of value: operational value for Biosphere 2, mechanistic and integrative value for soil and critical-zone science, methodological value for other observatories studying emergent multiscale behavior, and educational value through research-integrated mathematical-modeling training.

18 Physical Meaning of the Expected Results

The output of the proposed project should be stated in physical variables, not only in abstract geometric indices. The following quantities constitute the principal physical results.

1. **Effective gas diffusivity D_{eff} .** From the vertical phase gradient of a harmonic or narrow-band component,

$$D_{\text{eff}} \approx \frac{\omega}{2(\partial_z \phi)^2} \quad (111)$$

in the weak-sink limit, or from the joint amplitude–phase relation

$$D_{\text{eff}} = \frac{\omega}{2\alpha\beta} \quad (112)$$

when attenuation and phase delay are both resolved.

Physically, D_{eff} summarizes the ability of CO_2 to propagate through the gas-filled pore network. Long-term decreases can indicate reduced gas connectivity caused by greater water occupancy, increased tortuosity, pore restructuring, or accumulation of secondary material. Long-term increases could indicate drying, creation of preferential gas pathways, or structural reorganization that improves connectivity. Because D_{eff} is an effective parameter, independent moisture and porosity information is required before assigning a unique microscopic cause.

2. **Effective reaction/sink coefficient κ_{eff} .** For

$$q = \alpha + i\beta, \quad q^2 = \frac{\kappa + i\omega}{D}, \quad (113)$$

the local estimate is

$$\kappa_{\text{eff}} = D_{\text{eff}}(\alpha^2 - \beta^2). \quad (114)$$

This apparent parameter measures attenuation or exchange of the gas-phase CO_2 signal on the selected time scale within the restricted model. It may combine dissolution into water, aqueous reaction, changing gas storage, unrepresented transport, and boundary effects; it is not by itself a weathering rate or permanent carbon-removal estimate. LEO studies of pore-space CO_2 depletion and hydrologically dependent carbon uptake motivate testing this interpretation [5, 3].

A systematic change in κ_{eff} at matched moisture and forcing would therefore be a candidate change in apparent gas-signal removal/exchange. Assigning it to increasing reactive capacity requires aqueous/solid mass balance and chemical sampling.

3. Unresolved balance residual S_{res} . After estimating represented storage, transport, and exchange terms, the residual balance is

$$S_{\text{res}} \simeq \frac{\partial C}{\partial t} - \nabla \cdot (D_{\text{eff}} \nabla C) + \kappa_{\text{eff}} C + \nabla \cdot (\mathbf{v}_g C). \quad (115)$$

It collects unresolved source, storage, phase-exchange, transport, boundary, and model-error contributions and is not a direct microbial respiration measurement. Only after independent constraints close the non-biological terms may covariation with microbial biomass or respiration support an estimate of biological contribution.

4. Diffusive penetration depth and response time. The frequency-dependent penetration depth

$$\delta(\omega) = \sqrt{\frac{2D_{\text{eff}}}{\omega}} \quad (116)$$

and the corresponding diffusion time

$$\tau_D \sim \frac{L^2}{D_{\text{eff}}} \quad (117)$$

give directly interpretable time and length scales. These quantities indicate how deeply a surface or near-surface CO_2 perturbation can influence the basalt profile on a given time scale. A shift in τ_D over years is a direct physical measure of changing transport organization.

5. Structural model dimension q_* and normalized width η_q . These are empirical representation quantities, not material parameters or counts of mechanisms. They quantify how many affine directions are required to reproduce the selected observations within a prescribed error under a declared embedding. A rise in q_* means only that the previous low-order representation no longer captures those observations. Additional coupling or heterogeneity is one candidate explanation to test against forcing, sensor, and direct soil-state alternatives.

6. Grassmannian displacement of the optimal response subspace. The distance

$$d_G(L_q^*(t_1), L_q^*(t_2)) \quad (118)$$

measures reorientation of the selected empirical response subspace. A large value with nearly unchanged q_* means that representation dimension is similar while orientation changes. This can screen for functional reorganization not visible in overall variance, but geometry alone neither identifies the physical mixture nor establishes soil evolution.

7. Rain-event structural relaxation time. The width-recovery time

$$\tau_W \quad (119)$$

measures how rapidly the multidimensional response returns to its pre-event organization after rainfall. It can be compared with water-storage relaxation, discharge transit time, and τ_D . If τ_W becomes systematically longer during landscape evolution, the result would indicate increasing memory or coupling among hydrological, chemical, and biological processes.

8. Geometric hysteresis γ_{cycle} . The rain–dry geometric phase does not represent a conventional material coefficient. It is an integrated measure of path dependence. A nonzero and reproducible γ_{cycle} means that wetting and drying do not traverse the same sequence of normalized CO_2 states. Physically, this can arise from delayed water redistribution, reaction kinetics, microbial response, and irreversible or slowly reversible changes in pore and mineral structure.

9. Connection to independently measurable weathering and carbon quantities. The strongest version of the project should explicitly test the inferred parameters against quantities measured by established LEO methods:

$$D_{\text{app}}, \kappa_{\text{app}} \leftrightarrow S_{\text{hydro}}, \text{HCO}_3^-, \text{discharge chemistry}, \quad (120)$$

$$S_{\text{res}} \leftrightarrow \text{independent respiration, microbial function, and non-biological balance terms}, \quad (121)$$

and

$$q_*, \eta_q, d_G, \gamma_{\text{cycle}} \leftrightarrow \text{mineralogical and colloidal evidence of increasing heterogeneity}. \quad (122)$$

This cross-validation is essential. The proposed approach does not claim that a CO_2 time series alone uniquely determines mineralogy or microbial ecology. Its value is that it converts the dense sensor record into time-resolved, model-admissible apparent parameters and empirical response descriptors that can be tested against direct geochemical, hydrological, and biological observations.

19 Scientific Contribution, Innovation, and Project Value

The project combines established mathematical ingredients with new, application-specific links that still require validation. Multidimensional time-series unfolding and its Grassmannian formulation were developed by Buchstaber [13, 15]; geometric-phase methods have a long theoretical history [19, 20, 21]; and the restricted physical inversion is derived from a standard reaction–diffusion equation. Their integration for multiphase LEO pore-gas data is not established by those sources and is the hypothesis being tested.

Buchstaber’s unfolding theory contributes a precise language for affine representation, recurrence, and Grassmannian change. In this project, its computable squared-energy width $\mathcal{W}_q^{(2)}$ asks how accurately selected lag vectors can be represented by a q -dimensional affine subspace under stated sampling, centering, and loss conventions. It remains distinct from the classical worst-case Kolmogorov width d_q and, after normalization, is the PCA residual-variance curve. A changed profile is therefore an empirical response change at the analyzed scale, not a physical state dimension or proof of pedogenesis. The full definitions and limitations are developed in the preceding mathematical sections.

The innovation lies in experimentally testing the integration, not in claiming that any component alone identifies pedogenesis. The three descriptions remain distinct: unfolding/Grassmann metrics quantify empirical response geometry; a restricted transfer-function inversion estimates apparent transport–exchange parameters when its assumptions pass; and projective geometry describes normalized path orientation. Their interpretation requires independent slow-state and mass-balance measurements. The larger project value is operational: a successful framework will turn controlled events into auditable evidence, conditional predictions, and better choices of the next experiment even if no topological phenomenon is found.

Contribution 1: the state of LEO is represented by an evolving transport–reaction operator together with an evolving data-driven subspace. Conventional environmental time-series analysis normally treats each measured variable as a signal to be filtered, decomposed, or regressed. Here the operative state is richer:

$$[\mathcal{L}(t), L_q^*(t), |\Psi_B(t))]. \quad (123)$$

The first element is physical, the second is structural, and the third is geometric. The project therefore distinguishes changes of parameters *within* a dynamical organization from changes of the dynamical organization itself.

Contribution 2: unfolding geometry becomes an empirical screening and regularization diagnostic. The project uses centered unfolding geometry to identify reproducible, compressible response windows and to detect departure from baseline. This may improve numerical regularity but does not establish physical validity. The reaction–transport model is accepted only after separate tests of boundary conditions, coherence, conservation residual, parameter identifiability, and held-out prediction.

Contribution 3: rank-preserving continuation is converted into an empirical change-point test. Buchstaber gives exact recurrence and rank-preserving extension conditions for exact finite-rank series. In noisy LEO data, cross-validated recurrence failure is only a project-specific change statistic. Aligning it with rain and apparent parameters makes it a candidate for mechanistic follow-up, not proof of a physical regime change.

Contribution 4: two geometries are analyzed simultaneously. The project follows both

$$L_q^*(t) \in G(n, q) \quad (124)$$

and

$$|\Psi_B(t)\rangle \in \mathbb{CP}^{q-1}. \quad (125)$$

The Grassmann path measures reorientation of the low-dimensional dynamical subspace, while the projective path measures phase organization inside that subspace. This separation is valuable because the two kinds of reorganization need not occur together.

Contribution 5: controlled rain pulses provide repeated perturbation experiments for geometric time-series diagnostics. Buchstaber’s atmospheric-CO₂ example demonstrates the ability of unfolding-based approximations to separate trend and cyclic structure, but it is observational. LEO adds controlled perturbations and replicated hillslopes. This allows repeated tests of whether the same forcing produces the same path in structural and geometric state space. The combination of controlled forcing, physical inversion, unfolding geometry, and geometric phase is the distinctive experimental feature of the proposed project.

Contribution 6: long-term basalt-to-soil evolution is formulated as drift in dynamical organization rather than only drift in concentrations. The proposed end point is not a new concentration index. It is a quantitative description of how the empirical response subspace, model-admissible apparent transport–exchange parameters, and geometric hysteresis change together over years. This makes the framework suitable for detecting candidate functional reorganization of an evolving reactive substrate, to be interpreted only with independent slow-state evidence.

Contribution 7: the analysis closes the loop from observation to decision. Uncertainty-aware AI is used to emulate declared event responses, expose out-of-domain conditions, and rank feasible perturbations or samples by expected information gain. The contribution is not a black-box predictor but a prospectively testable experiment–model–data cycle in which failed predictions and null results update both the physical hypotheses and the next design.

20 Advantages Relative to Existing Analysis Strategies

The principal advantage of the proposed framework is not that it replaces established methods, but that it resolves ambiguities left by each method when used alone.

Method used alone	Main limitation for LEO	Added capability of the proposed framework
Mean concentration, trends, or anomalies	Different physical mechanisms can produce similar changes in C .	Restricted amplitude–phase inversion adds complementary constraints on transport and exchange and provides an unresolved balance residual, with explicit admissibility flags.
Fourier or wavelet spectra	Identify characteristic frequencies but do not determine whether a peak corresponds to a stable low-dimensional dynamical structure.	Buchstaber unfolding tests whether an oscillatory component is supported by a stable approximately rank-two geometry and follows its subspace through time.
PCA/SSA/Hankel decomposition	Efficiently reduces dimension, but a singular-value spectrum alone does not distinguish subspace rotation from changes of modal energy.	The optimal subspace is treated as a point on $G(n, q)$; both q -width and Grassmannian motion are tracked.
Generic change-point detection	Detects statistical change but often gives little mechanistic interpretation.	Rank-preserving continuation gives an empirical change statistic that can be compared with rain forcing, admissible D_{app} , κ_{app} , and S_{res} .
Reaction–diffusion parameter fitting	Can be ill-conditioned and can produce parameters even when the assumed low-dimensional model is not supported by the data.	Unfolding rank and q -width are used as pre-fit admissibility tests, reducing interpretation of structurally inappropriate windows.
Geometric phase calculated directly from raw sensors	Sensitive to noise, sensor redundancy, and arbitrary channel composition.	Geometric phase is calculated in a low-dimensional subspace whose stability has first been validated from the unfolding.
Topological diagnostics applied directly to a complex field	Can produce visually compelling but fragile defects when amplitudes are small or interpolation is unstable.	Topology is explicitly placed at the final evidence level, after structural, physical, and geometric validation.

Improved interpretability. Every major diagnostic has a distinct role:

$$\mathcal{W}_q^{(2)} \rightarrow \text{affine representation error at scale } q, \quad (126)$$

$$d_G \rightarrow \text{rotation of the selected empirical subspace}, \quad (127)$$

$$E_{\text{cont}} \rightarrow \text{failure of the fitted recurrence}, \quad (128)$$

$$(D_{\text{app}}, \kappa_{\text{app}}, S_{\text{res}}) \rightarrow \text{candidate physical constraints}, \quad (129)$$

$$(\gamma_g, d_{\text{FS}}) \rightarrow \text{normalized path descriptors}. \quad (130)$$

This separation prevents several mathematically different phenomena from being compressed into one generic “complexity” score.

Improved robustness. The framework supplies several internal consistency checks. The same event can be required to appear simultaneously as a change in width, recurrence residual, subspace orientation, admissible apparent parameters, and geometric trajectory. Single-diagnostic anomalies can therefore be screened for additional scrutiny, although coincident descriptors are not independent proof of mechanism.

Improved sensitivity to reorganization without large amplitude change. Buchstaber’s approach is especially useful when the dominant signal amplitude remains similar but the geometry of the unfolding changes. Deymier-style geometric phase has the analogous advantage in complex state space. Their combination is therefore well suited to the central LEO hypothesis that the substrate can reorganize dynamically before a large change becomes obvious in mean CO_2 .

Improved use of the experimental design of LEO. Many time-series methods are designed for passive observations. The proposed analysis explicitly uses controlled rain events and the three replicate hillslopes as repeated perturbation experiments. This makes it possible to test reproducibility and estimate the response to assigned rainfall treatments under comparable antecedent conditions more directly than in a single uncontrolled record. It does not causally identify biological versus geochemical pathways.

A falsifiable contribution claim. If hydrologically and thermally matched rain events produce stable $\mathcal{W}_q^{(2)}$, L_q^* , admissible D_{app} , κ_{app} , and γ_g across years, then these observables provide no evidence for long-term response reorganization. Conversely, systematic drift across these levels establishes change in the measured functional response only after sensor, forcing, and model-discrepancy alternatives are screened. A soil-development interpretation additionally requires independent mineralogical, hydraulic, chemical, and biological evidence. This remains useful even when the result is null.

21 Key Integration: Geometric Phase from Robust Modes, Not Only Raw Channels

Direct analysis of $|\Psi\rangle$ constructed from all sensor channels is useful as a reference, but it may be sensitive to sensor noise, drift, and redundant dimensionality. The principal integrated strategy is therefore to first identify a low-dimensional dynamical subspace.

Block-Hankel Representation For M sensors, construct trajectory matrices $H_1, \dots, H_M$. A natural multichannel representation is

$$H_B = \begin{pmatrix} H_1 \\ H_2 \\ \vdots \\ H_M \end{pmatrix}, \quad H_B = U_B \Sigma_B V_B^*. \quad (131)$$

The SVD extracts dominant modes of the stacked matrix; it does not prove that they are shared across sensors. Each channel must first be scaled using a declared noise/variance convention, and cross-sensor loadings, coherence, and leave-one-channel-out stability must distinguish shared from high-energy sensor-specific structure.

Complex Modal Coefficients For the leading r modes, compute complex amplitudes

$$a_k(\omega, t) = A_k(\omega, t) e^{i\vartheta_k(\omega, t)} \quad (132)$$

using either a Fourier transform or an analytic-signal/Hilbert representation of the modal time coefficients. Then define the modal state

$$|\Psi_B(\omega, t)\rangle = \frac{(a_1, \dots, a_r)^\top}{\sqrt{\sum_{k=1}^r |a_k|^2}}. \quad (133)$$

The geometric phase and d_{FS} are then computed in the space of robust dynamical modes.

Phase-Preserving Rank Ordinary rank selection based solely on reconstruction error may be inadequate because a mode with low energy can still substantially alter the complex orientation of the state. Define

$$\Delta\gamma_r = \left| \text{wrap} \left[\gamma_g^{(r)} - \gamma_g^{(\text{full})} \right] \right|. \quad (134)$$

Then

$$r_\phi = \min \{r : R_r < \varepsilon_R, \quad \Delta\gamma_r < \varepsilon_\gamma\}. \quad (135)$$

The quantity r_ϕ is the minimum rank that preserves both the amplitude structure and the trajectory geometry.

22 Synergy with Physical Inversion

Low-rank analysis does not replace the physical model; rather, it helps identify *where* the physical model is reliable.

If, in a given window,

1. R_r is small;
2. r_{eff} is stable under reasonable changes in L ;
3. the phase lag remains stable under changes in window width and sensor subset,

then a simple harmonic reaction–diffusion fit has a much clearer physical interpretation.

For a pair of depths $z_1 < z_2$,

$$\beta_{12} \simeq -\frac{\phi_2 - \phi_1}{z_2 - z_1}, \quad \alpha_{12} \simeq -\frac{\ln |\hat{C}_2/\hat{C}_1|}{z_2 - z_1}. \quad (136)$$

With three levels, self-consistency can be tested by comparing

$$(\alpha_{12}, \beta_{12}) \quad \text{and} \quad (\alpha_{23}, \beta_{23}). \quad (137)$$

Large discrepancies indicate that the homogeneous one-dimensional approximation may be breaking down because of a local source/sink, advection, or spatial heterogeneity.

Once model-admissible apparent D and κ are available, an unresolved balance residual may be estimated from

$$S_{\text{res}} \simeq \frac{\partial C}{\partial t} - \nabla \cdot (D \nabla C) + \kappa C + \nabla \cdot (\mathbf{v}_g C), \quad (138)$$

This quantity collects unresolved source, storage, exchange, advection, boundary, and model-error terms and must not be interpreted as microbial production without independent closure.

23 Spatial Level: From Vertical Columns to the Hillslope

Let the vertical columns be located at (x_m, y_m) . For each column, compute

$$V_1^{(m)}, R_r^{(m)}, r_{\text{eff}}^{(m)}, \gamma_g^{(m)}, D^{(m)}, \kappa^{(m)}. \quad (139)$$

This yields spatial maps

$$V_1(x, y, t), \quad r_{\text{eff}}(x, y, t), \quad \gamma_g(x, y, t), \quad D_{\text{eff}}(x, y, t), \quad \kappa_{\text{eff}}(x, y, t). \quad (140)$$

Between columns one may compute

$$d_{\text{FS}}(m, n) = \arccos |\langle \Psi_m | \Psi_n \rangle|, \quad (141)$$

and compare this projective-state distance with physical distance, slope position, and known hydrological pathways.

The three LEO hillslopes are particularly valuable as replicate systems [1]. One can therefore investigate

$$d_{\text{FS}}(E, C), \quad d_{\text{FS}}(E, W), \quad \Delta\gamma_{EC}, \quad \Delta\gamma_{EW}, \quad (142)$$

and test whether initially similar systems diverge in projective state space as biological and geochemical evolution proceeds.

24 Frequency and Time Scales

For a diffusion process, the characteristic penetration depth is

$$\delta(\omega) \sim \sqrt{\frac{2D}{\omega}}. \quad (143)$$

For a characteristic depth $L_z \sim 1$ m, the corresponding diffusion time is

$$\tau_D \sim \frac{L_z^2}{D}. \quad (144)$$

If $D \sim 10^{-6} \text{ m}^2 \text{ s}^{-1}$, then $\tau_D \sim 10^6$ s, i.e., on the order of days to weeks. This is not asserted as a universal value for LEO; it is a scale estimate showing that informative bands may lie not only at daily frequencies but also on multi-day and weekly scales.

A practical multiscale analysis is:

Scale	Possible physical process	Recommended analysis
12–24 h	diurnal forcing, thermal modulation	Fourier phase + short-window Hankel
2–10 d	rain response, redistribution	Buchstaber/SSA metrics + γ_g
1–6 weeks	slow diffusion / biological response	multiscale Hankel, modal state
season	structural wet–dry hysteresis	closed-cycle geometric phase
year+	landscape evolution	drift of $r_{\text{eff}}, d_{\text{FS}}, \gamma_g, D, \kappa$

25 Geometric Phase Versus Genuine Topology

Geometric diagnostics and topological invariants must be distinguished rigorously.

Level	What can be claimed	Required validation
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A	Structural complexity	robust Hankel/SVD metrics under changes in window and lag length
B	Geometric evolution	stable γ_g and d_{FS} , with gauge consistency
C	Phase defect	robust amplitude zero/singularity and integer winding
D	Non-Hermitian topology	reconstructed operator and reproducible spectral branching/EP-like behavior

Spatial Phase Winding For a complex response field

$$Z(x, y, \omega) = A(x, y, \omega) e^{i\phi(x, y, \omega)}, \quad (145)$$

one may compute around a closed spatial contour $\mathcal{C}$

$$W = \frac{1}{2\pi} \oint_{\mathcal{C}} \nabla \phi \cdot d\mathbf{l}. \quad (146)$$

A nonzero W has physical meaning only if:

1. the contour encloses a robust, reproducible low-amplitude region where phase can become undefined;
2. the integer result survives changes in interpolation, window, frequency band, and sensor subset;
3. the defect cannot be attributed to a single malfunctioning sensor.

Such an object would not represent a hydrodynamic vortex of the gas. It would represent a topological defect of the complex response field.

Non-Hermitian Extension A real diffusion–reaction operator can be self-adjoint under standard Dirichlet, Neumann, or suitable Robin conditions; dissipation or a reaction sink alone does not make it non-Hermitian. Advection, asymmetric coupling or boundary conditions, and data-driven nonnormal reductions may make the operator non-self-adjoint under a stated inner product. Only after that property is established should one use left and right eigenvectors:

$$\mathcal{L} |R_n\rangle = \mu_n |R_n\rangle, \quad \langle L_n | \mathcal{L} = \mu_n \langle L_n |, \quad \langle L_n | R_m \rangle = \delta_{nm}. \quad (147)$$

For non-Hermitian systems, geometric phase requires a biorthogonal formulation; such generalizations were developed, for example, by Garrison and Wright [23]. One may define

$$\mathcal{A}_n(\lambda) = i \langle L_n | \nabla_\lambda R_n \rangle. \quad (148)$$

Exceptional points are characteristic of non-Hermitian spectral problems: eigenvalues and eigenvectors can coalesce, and encircling such degeneracies can produce nontrivial state permutation and geometric structure [24]. Their existence in LEO should not be assumed. A practical route would first reconstruct a data-driven reduced operator, for example by DMD, Koopman approximation, or state-space identification, and only then examine its spectrum for reproducible evidence of near-degeneracies, branch exchange, or winding.

26 New Integral Diagnostics

The following quantities are proposed as *diagnostic indices*, not as fundamental invariants.

Geometric Complexity Index Define

$$\text{GCI}(t) = r_{\text{eff}}(t) \frac{|\gamma_g(t)|}{\pi}. \quad (149)$$

This quantity is large when the trajectory is simultaneously multimodal and characterized by a pronounced oriented geometric phase.

Modal Redistribution Index If

$$p_k(t) = \frac{\sigma_k^2(t)}{\sum_j \sigma_j^2(t)}, \quad (150)$$

define

$$\text{MRI}(t_1, t_2) = \frac{1}{2} \sum_k |p_k(t_2) - p_k(t_1)|. \quad (151)$$

MRI quantifies redistribution of modal energy. In combination with d_{FS} , it distinguishes cases in which only the singular-value spectrum changes from cases in which the complex orientation of the state changes as well.

Phase-Preserving Rank The quantity r_ϕ defined in (135) is particularly important because it measures how many modes are required not merely for reconstruction but for preservation of phase geometry.

27 Data-Driven Prediction and Experiment–Model Feedback

The AI component is a specified, testable work package rather than a general claim that machine learning will discover mechanism. Its primary role is to learn conditional event-response relationships, quantify uncertainty and distribution shift, and support the choice of the next informative experiment or sample.

Declared prediction problem. For event e , let the inputs contain the rainfall or boundary-forcing history $\mathbf{u}_e$, antecedent moisture, temperature, storage and pressure state $\mathbf{h}_e$, sensor-quality metadata $\mathbf{q}_e$, and available slow-state observations $\boldsymbol{\xi}_e$. A predictor $\mathcal{M}$ is trained for a declared target $\mathbf{y}_e$:

$$\hat{\mathbf{y}}_e, \mathcal{U}_e = \mathcal{M}(\mathbf{u}_e, \mathbf{h}_e, \mathbf{q}_e, \boldsymbol{\xi}_e), \quad (152)$$

where $\mathbf{y}_e$ may be a complete depth-resolved event trajectory or a predeclared set of response descriptors and $\mathcal{U}_e$ is calibrated predictive uncertainty. The target, forecast horizon, preprocessing, and admissible operating domain must be fixed before evaluation.

Validation design and baselines. Samples from one event must not be split between training and testing. Entire events and, for temporal transfer, entire years are held out. Leave-one-hillslope-out tests are reported as exploratory stress tests rather than independent population validation. AI models are compared with persistence, climatological or matched-event averages, conventional regression, linear state-space models, and physically restricted reduced-order models. Performance includes trajectory or descriptor error, predictive-interval coverage and sharpness, calibration, conservation residual, sensor-removal sensitivity, and an explicit abstention rate under out-of-distribution forcing or state.

Physics and geometry as constraints. The emulator may use physical residuals, monotonicity or positivity constraints where justified, model-admissibility flags, and robust mathematical descriptors as inputs or auxiliary targets. It may not replace missing mineralogical or biological evidence with a latent variable and call that variable a mechanism. Disagreement among a physical model, geometric descriptors, and an AI predictor is retained as a useful discrepancy signal rather than averaged away.

Active experimental design. For a finite set of feasible candidate perturbations or sampling plans $a \in \mathcal{A}$, the design module ranks actions by a transparent utility such as expected reduction in parameter or model uncertainty, discrimination among candidate hypotheses, coverage of poorly observed state space, operational cost, and experimental risk:

$$a^* = \arg \max_{a \in \mathcal{A}} [\mathbb{E}\{\Delta I(a)\} - \lambda_C C(a) - \lambda_R R(a)]. \quad (153)$$

At least one ranked design action should be tested prospectively. The loop is considered successful only if the selected action reduces uncertainty or discriminates hypotheses better than a predeclared standard design.

28 Phase III of the Project: Experimental-Design Learning and Prospective Validation

Phase III converts the scientific limitations found in Phases I–II into an additional result of the research: quantitative knowledge about how the LEO experiment can become more informative. This is broader than asking an ML model to choose one more rain event. Physical non-identifiability, sensor geometry, missing phase measurements, sampling mismatch, and operational experience may be more important than predictive uncertainty. Experimental-design learning therefore stands alongside physical inference, mathematical structure, and prediction as a project outcome in its own right.

28.1 Experimental Design as a Scientific and Operational Result

The present LEO design has already enabled exceptional controlled Earth-system research. Learning how it can evolve is not a claim that the original design was deficient; it is a natural result for an observatory whose long record reveals new spatial gradients, time scales, blind spots, and scientific opportunities. Biosphere 2 leadership, LEO investigators, and observatory operators become co-users of this result. Every suggested improvement must state the supporting evidence, uncertainty, operational cost, risk, expected scientific gain, and effect on continuity of the long-term record.

1. Replace opportunistic event comparison with a planned forcing library. Define a manageable matrix of rainfall intensity, duration, total input, spatial delivery, antecedent moisture/storage, temperature regime, and dry-down duration. A complete factorial design is neither necessary nor operationally realistic. Instead, use blocked and sequential designs: first repeat matched reference events to estimate repeatability, then change one high-value forcing dimension at a time or use an information-based design within safety and water constraints. Every experimental event should have a declared purpose, forcing-tolerance envelope, baseline comparator, and recovery window.

2. Reassess sensor depths and locations through observability, not convenience alone. Sensitivity, coherence, profile-curvature, leave-one-sensor-out, and posterior-uncertainty analyses should identify depth intervals and hillslope positions where the present network cannot distinguish transport, storage, exchange, or boundary effects. The resulting observability map should support a ranked choice among retaining, adding, relocating, or redundantly pairing sensors. Candidate

additions include atmospheric and basal references, gas pressure, gas-filled porosity or its components, oxygen where scientifically justified, and redundant CO₂ measurements at high-information depth transitions. Any decision to move or add hardware must be based on prospective uncertainty reduction and must respect the continuity value of the long-term record.

3. Establish co-located gas–water–solid–biology validation hubs. A small number of intensively characterized sentinel locations may be more informative than sparse expansion of every variable across the entire hillslope. At these hubs, pore-gas CO₂, moisture, temperature, pressure, surface exchange, pore-water DIC/alkalinity and major solutes, mineral/colloid observations, organic carbon, and biological measurements should be synchronized as far as feasible. Co-location is essential for determining whether a gas-phase residual is compatible with phase exchange, aqueous export, mineral reaction, or biological production. The exact measurement set should be selected jointly with domain specialists and should not be inferred from the CO₂ analysis alone.

4. Introduce trigger-based direct sampling. Fixed calendar sampling can miss short-lived wetting fronts, exchange pulses, and transitions identified by continuous sensors. The response atlas should therefore generate preregistered triggers—for example, an admissible change in vertical lag, a persistent model discrepancy, or an uncertainty-qualified departure from a matched-event envelope. A trigger should activate an agreed sequence of aqueous, gas-flux, mineral/colloid, or biological samples at selected sentinel sites during onset, peak response, and recovery. False-alert limits, staff availability, preservation requirements, and safe operating windows must be set before deployment.

5. Strengthen controls, staggering, and hierarchical replication. The three hillslopes are valuable matched systems but only limited system-level replication. Where operations permit, future treatments should use staggered timing, matched reference events, randomized treatment order, or temporary within-hillslope controls so that treatment response can be distinguished from common atmospheric forcing and long-term drift. The design should preserve the value of coordinated hillslope experiments while creating explicit counterfactual comparisons when the scientific question requires them.

6. Make calibration and failure tests part of the experiment. Reference-gas checks, cross-sensor comparisons, maintenance metadata, pressure diagnostics, and planned redundancy should be scheduled as experimental controls rather than treated only as post hoc quality assurance. Events intended for physical inversion should include pre-event and post-event checks sufficient to distinguish landscape response from sensor response. This experimental-design analysis should identify which apparent scientific uncertainties are actually calibration or metadata uncertainties.

7. Expand the experiment where model disagreement is informative. Regions in which the physical inversion, unfolding geometry, direct samples, and AI emulator disagree are not merely failed analyses. They identify candidate regimes for deeper investigation. Phase III should rank such regimes by scientific importance, reversibility, feasibility, and ability to distinguish competing hypotheses. Possible responses include new antecedent-moisture regimes, longer monitored dry-downs, denser sampling around a depth transition, or targeted direct measurements rather than automatically increasing model complexity.

28.2 Evidence-to-Design Decision Matrix

Each identified design opportunity should trace an observed limitation to a candidate change and a prospective test:

Evidence from analysis	Candidate design response	Prospective success test
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Poor vertical identifiability or high sensitivity to one sensor	Add, relocate, or redundantly pair a sensor at a high-information depth	Reduced posterior uncertainty and leave-one-sensor sensitivity
Apparent κ confounded with storage or gas–water exchange	Add co-located pressure, gas-filled-porosity, aqueous DIC/alkalinity, and flux constraints	Improved balance closure and rejection of at least one candidate model
Nominally similar rain events are not dynamically comparable	Introduce reference rainfall recipes and antecedent-state acceptance windows	Lower forcing/state mismatch and narrower reference-response envelope
Short-lived anomaly lacks direct mechanistic observations	Activate trigger-based water, solid/colloid, flux, or biological sampling	Direct sample obtained within the predeclared response window
AI emulator abstains in a scientifically important regime	Schedule a safe perturbation that expands coverage of that regime	Improved calibration or reduced epistemic uncertainty on a later held-out event
Hillslope responses diverge without a counterfactual	Stagger treatment timing or retain a matched reference hillslope/event	Better separation of common forcing, treatment response, and baseline drift
Apparent change follows maintenance or calibration history	Add reference checks, redundancy, and prospective maintenance controls	Artifact reproduced or excluded without invoking landscape change

28.3 Tiered Experimental-Design Outcomes

To make this experimental-design result usable under different budgets, Phase III will organize the identified opportunities into three tiers:

1. **Tier A—protocol and data improvements:** common event definitions, forcing recipes, antecedent-state gates, preregistered exclusions, calibration schedules, metadata standards, sample timing, and decision dashboards. These changes primarily use existing infrastructure.
2. **Tier B—targeted measurement augmentation:** sentinel validation hubs, limited redundant sensors, pressure or phase-exchange constraints, and trigger-ready sampling resources at locations selected by observability analysis.
3. **Tier C—strategic infrastructure or manipulation:** relocation or addition of high-value sensor profiles, new controlled forcing regimes, longer recovery experiments, or carefully selected interventions that require substantial operational review.

For every tier, the project will report expected information gain, scientific question addressed, cost class, operational burden, reversibility, risks to long-term continuity, and the evidence needed to discontinue an unproductive change.

28.4 Integration with the LEO Decision Cycle and Criterion for Completion

The scientific result enters practice through an annual or campaign-level experimental-design review involving Biosphere 2 leadership, observatory operators, hydrologists, geochemists, soil and microbial scientists, sensor specialists, and the modeling team. The project contributes the sensor-failure calendar, information-gap map, ranked design opportunities, tiered costs, dependencies, uncertainty, and a proposed prospective test. Leadership and operators contribute feasibility, safety, continuity, and scheduling constraints. The result is therefore jointly evaluated evidence about the experiment rather than an automated command.

Phase III is complete when at least one feasible protocol, sampling, measurement, or forcing change has been selected through this review, preregistered, implemented, and compared with a

standard design. Success means demonstrable improvement in identifiability, uncertainty calibration, response coverage, or hypothesis discrimination. Failure to improve is also informative if it is documented, because it prevents unsupported expansion and redirects the next design cycle.

29 Statistical Validation and Protection Against False Positives

Geometric phase can be highly sensitive; for that reason it requires stricter validation than a conventional amplitude statistic.

A minimal validation framework should include:

1. computer-algebra verification of every inverse identity and unit test of limiting cases;
2. synthetic forward-inverse tests for a semi-infinite profile, a finite column with both boundaries, distributed sources, variable saturation, noise, and sensor bias; the simple inversion must recover truth only in its declared regime and reject the others;
3. unit tests of centered affine rank, tail-eigenvalue projection error, anti-diagonal reconstruction, degenerate sinusoidal sampling, and eigengap nonuniqueness;
4. repeating the analysis for several lag lengths L ;
5. repeating it for several sliding-window widths;
6. bootstrap resampling across temporal windows and sensor channels;
7. leave-one-sensor-out sensitivity analysis;
8. explicit separation of reliable and questionable sensor periods;
9. parallel analysis of raw amplitudes and normalized states;
10. surrogate time series that preserve the power spectrum but destroy phase relations;
11. testing the stability of γ_g with respect to rank r ;
12. for winding numbers, testing stability with respect to frequency band, interpolation, and contour choice;
13. performing interannual comparisons only at comparable hydrological states, so that seasonal forcing is not mistaken for landscape evolution.

This last point is particularly important because the recent hillslope-scale basalt-weathering experiment in LEO demonstrated effects of infiltration and water saturation on P_{CO_2} and carbon uptake [3]. Hydrological state should therefore be treated as a conditioning variable rather than as a nuisance covariate.

30 Integrated Analysis, Experimental-Learning, and Decision Workflow

1. **Sensor forensics and failure calendar.** Begin from the completed 144-channel audit; update dated liveness, degradation, failure, maintenance, and recovery intervals without modifying raw data, and issue both analysis masks and operator-facing flags.
2. **Synchronization.** Place only admissible CO_2 intervals, temperature, moisture, atmospheric CO_2 , rain/irrigation, water storage, and sensor-quality metadata on a common time base.
3. **Physically meaningful windows.** Identify rain-response intervals, dry periods, seasonal windows, and comparable interannual states.

4. **Lag embedding.** Construct node-oriented Hankel matrices at several values of L , then center lag nodes explicitly for Buchstaber affine geometry; retain raw-Hankel/SSA quantities under separate names.
5. **Structural metrics.** Compute eigenvalues, $\mathcal{W}_q^{(2)}$, η_q , eigengaps, bootstrap projector uncertainty, and optional entropy rank, evaluating sensitivity to sampling, centering, window, L , and noise.
6. **Block-Hankel analysis.** Construct multichannel representations for vertical columns, then, if justified, for groups of columns and for an entire hillslope.
7. **Representation selection.** Choose r from noise-calibrated held-out reconstruction, eigengap stability, and sensitivity analysis. Treat $\Delta\gamma_r$ as a secondary geometric convergence check, not a physical rank criterion.
8. **Complex modal amplitudes.** Within selected frequency bands, compute $a_k(\omega, t)$, then construct $|\Psi_B\rangle$.
9. **Geometric diagnostics.** Compute d_{FS} , the Pancharatnam phase, and the closed-cycle γ_{cycle} .
10. **Vertical inversion.** Only where coherence, finite-domain boundary conditions, source tests, and profile fit are adequate, estimate $\alpha, \beta, D_{\text{app}}, \kappa_{\text{app}}$ with full uncertainty; otherwise label the window model-inadmissible.
11. **Unresolved balance residual.** Estimate S_{res} only by regularized state/weak-form inversion, never relabeling it biological without independent constraints.
12. **Spatial maps.** Map empirical metrics and admissible apparent parameters with interval support and uncertainty; do not extrapolate sensor intervals to hillslope fields without a spatial observation model.
13. **Independent slow-state validation.** Test event descriptors against mineralogical, aqueous/solid mass-balance, pore-structure, water-retention, organic-carbon, and biological observations, including hydrology-, temperature-, and sensor-age null models.
14. **Hierarchical hillslope comparison.** Compare East, Center, and West while treating sensors and events as nested observations and the three hillslopes as limited system-level replication.
15. **AI prediction and uncertainty.** Train only on declared event groups; evaluate complete held-out events and years against simple and physical baselines; calibrate uncertainty and out-of-distribution abstention.
16. **Experimental-design learning.** Combine the failure calendar, observability map, physical non-identifiability, model discrepancy, direct-sample needs, and AI uncertainty into ranked opportunities to deepen or broaden the experiment. Record the selected change before execution and test whether it improves hypothesis discrimination, uncertainty, response coverage, or operational reliability.
17. **Research–education reproducibility loop.** Use quality-controlled subsets and clearly scoped problems in MATH 485/585; independently review and reproduce course-derived analyses before any result, code, or hypothesis enters the scientific pipeline.
18. **Reproducible release and transfer test.** Publish the failure calendar, event catalogue, schemas, software, synthetic benchmarks, educational case studies, and a deliberately shifted or external test with documented portability limits.
19. **Topology only at the final stage.** Only after all preceding validation should one search for robust W , phase singularities, or non-Hermitian spectral winding; these exploratory results are not required for project success.

31 Scientific Hypotheses

1. After matching forcing and hydrological/thermal state, the distribution of empirical response descriptors may drift with independently measured soil development. No sign is prescribed: affine width or entropy rank may increase or decrease.
2. Rain–dry hysteresis changes not only in amplitude but also geometrically; after conditioning on hydrological state, γ_{cycle} may serve as a sensitive interannual diagnostic.
3. Drift in γ_g may be partly explained by changes in admissible apparent D and κ ; residual geometric change identifies model discrepancy for follow-up, not a biological source by itself.
4. The three hillslopes may diverge in projective response space; hierarchical analysis can quantify that divergence while recognizing limited system-level replication.
5. Low-energy modes may be critical for geometric phase, so rank selection based only on variance or reconstruction error may be insufficient.
6. Physics-constrained and geometry-informed predictors may improve held-out event reproduction beyond persistence, matched-event averages, and conventional regression while retaining calibrated uncertainty and abstention under distribution shift.
7. A prospectively selected perturbation or sampling plan may reduce uncertainty or discriminate competing models more effectively than the predeclared standard design.

32 What Would Constitute the Most Convincing First Result

A concrete preliminary result already exists. The project has produced a reproducible 144-channel sensor audit and dated failure chronology, isolated a defensible cross-slope analysis window, and rejected direct interpretation of the 24-hour D1-to-D2 phase as downward gas diffusivity. This is valuable independently of any later geometric or AI result: it protects the science from failed sensors, gives the observatory actionable quality information, and identifies the measurements needed to resolve the remaining physics.

The strongest next integrative result would not be a declaration of a “topological transition.” It would be a reproducible, uncertainty-qualified chain across hydrologically and thermally matched controlled events:

$$\begin{aligned}
 \text{matched forcing and antecedent state} &\longrightarrow \text{coherent depth-resolved response} \\
 &\longrightarrow \{D_{\text{app}}, \kappa_{\text{app}}, S_{\text{res}}\} \\
 &\longrightarrow \text{stable structural and hysteresis descriptors} \\
 &\longrightarrow \text{held-out-event reproduction.}
 \end{aligned} \tag{154}$$

A systematic interannual shift in this chain would establish change in the measured functional response only after sensor drift, boundary forcing, pressure, temperature, antecedent hydrology, and model discrepancy are excluded. A basalt-to-soil interpretation would additionally require covariation with independent mineralogical, aqueous/solid mass-balance, hydraulic, organic-carbon, and biological observations.

A second strong result would be prospective early detection: a geometric or unfolding descriptor triggers direct sampling before a conventional indicator crosses its predeclared threshold, while maintaining a controlled false-alert rate. “Earlier” must be measured against a named comparator and sampling schedule, not asserted retrospectively.

A third strong result would be evidence-based improvement of the experiment itself. The failure calendar, physical identifiability analysis, geometry, direct-sample gaps, operational constraints, and AI uncertainty would jointly identify a feasible change in forcing, measurement, calibration, or sampling. The choice would be recorded before implementation and tested against a standard

design. A null improvement remains useful because it prevents unsupported expansion and redirects the next design cycle.

33 Overall Conceptual Framework

The integrated scheme may be summarized as

$$\begin{aligned}
\{\xi_e, \text{forcing, antecedent state}\} &\longrightarrow \{C_j, \theta_w, T, \text{storage, fluxes}\} \\
&\longrightarrow \{D_{\text{app}}, \kappa_{\text{app}}, S_{\text{res}}\} \\
&\quad \oplus \{\mathcal{W}_q^{(2)}, L_q^*, |\Psi_B\rangle, d_{\text{FS}}, \gamma_g\} \\
&\longrightarrow \{\text{uncertainty, discrepancy, competing hypotheses}\} \\
&\longrightarrow \{\text{AI prediction, observability gaps, operational evidence}\} \\
&\longrightarrow \text{experimental-design learning and prospective validation.}
\end{aligned} \tag{155}$$

Here ξ_e denotes independently observed slow soil state, while the other quantities describe event-scale response. The arrows do not assert unique mechanistic inversion: they define a sequence of constrained comparisons, falsification tests, and decisions.

The most scientifically meaningful formulation is therefore not to “apply topology to CO₂,” but to build a perturbation-based learning framework for an evolving basalt substrate. Sensor forensics protects the evidence base; Buchstaber-style unfolding supplies auditable structural coordinates; restricted physical inversion supplies model-dependent apparent transport–exchange constraints; Deymier-style geometry supplies path descriptors; AI supplies uncertainty-aware prediction; and the combined evidence reveals how the experiment itself can be made more informative. Topology remains optional and belongs only to a later evidence level.

34 Conclusion

The project operationalizes the basalt-to-soil question as a testable three-phase program: establish trustworthy perturbation-response observables; connect them to slow soil development with mathematics and uncertainty-aware AI; and use the resulting information gaps as knowledge about how the LEO experiment itself can improve. Its contribution is the complete learning cycle rather than any isolated index or model.

The work has already produced tangible value. A dense preliminary study audited all 144 basalt CO₂ channels, created a dated failure chronology and defensible analysis masks, isolated a reproducible cross-slope window, and rejected direct inversion of the 24-hour D1-to-D2 phase as gas diffusivity. Those results both protect subsequent science and supply operational signals about sensor status, metadata, calibration, and the need for co-located temperature, pressure, aqueous-carbon, and biological measurements.

For Biosphere 2, the continuing outputs will include the living sensor-failure calendar, event catalogue, response atlas, admissibility-aware inference and prediction software, sampling triggers, and evidence-based priorities for deepening or broadening forcing, measurement, and validation. Improvement of experimental design is therefore an additional scientific and practical result of the work, not merely an ML recommendation. For the broader community, the project will provide a documented benchmark and transferable protocol, with the limits of generalization stated explicitly.

The work also has an active educational life through its integration into MATH 485/585, *Mathematical Modeling*, in the University of Arizona Department of Mathematics. This creates a reviewed pathway from classroom derivation and reproducible computation to interdisciplinary research questions and validated observatory tools.

The guiding ambition remains to understand how mineral parent material acquires the organization and functions of a living, evolving soil system. The evidentiary standard remains deliberately stricter: a change in CO₂ response is evidence of changed function, while soil formation and its

causes require convergence across physical, chemical, mineralogical, hydrological, and biological observations.

Note on Preliminary Evidence and Reproducibility

The preliminary claims in this document are grounded in the existing B2 Inverse Dynamics research archive. The reproducibility bundle includes the full three-slope failure chronology, monthly liveness tables, the fixed 2017–2018 window manifest, machine-readable latency results, and scripts for synthetic and observational replay. Raw parquet data are read-only; quality decisions are stored as derived masks. The sensor-forensics result is separated from the more interpretive physics: failure dates and liveness can remain useful even if a later transport hypothesis is rejected. Established results, well-supported interpretations, provisional patterns, and open questions are maintained as separate evidence tiers.

Note on Citations

The bibliography includes only real sources whose contents are directly relevant to the statements they support. In particular, LEO papers are cited for the design and controllability of the observatory and for the experimentally measured relationship among P_{CO_2} , hydrology, and basalt weathering; SSA references are cited only for lag/Hankel/SVD methodology; Deymier et al. are cited only for the geometric-phase sensing analogy; Pancharatnam, Berry, and Samuel–Bhandari are cited for geometric-phase formalism; and Garrison–Wright and Heiss are cited for non-Hermitian geometric phase and exceptional-point physics. The composite indices proposed here (GCI, MRI, and r_ϕ) are constructions of the present document and are not attributed to those sources.

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